When Is It Too Late?

Causal Recoverability Curves for Long-Horizon Vision-Language-Action Manipulation

A hypothesis-driven LaTeX experiment plan for adapting SAFE, measuring actionable warning time, learning recovery value, and evaluating critic-guided recovery on RoboCasa365

Primary target:	CoRL
Natural alternative:	ICRA
Stretch target:	RSS
Conditional vision route:	CVPR / ICCV / WACV
Primary policy:	Xiaomi-Robotics-1 on RoboCasa365
Report date:	12 October 2026

This document starts from the SAFE and recovery evidence in the repository and incorporates results through 12 October 2026. Phase 1 frozen confirmation failed; Phase 2 complete snapshot replay passed; Phases 3 and 3B found negligible local-candidate selection headroom. The subsequent 160-suffix higher-level operator screen also passed engineering but failed its scientific gate: continuation reached 22.5% stage completion versus 17.5% for every intervention, with zero oracle routing gain and zero full-task success. The next proposed experiment is a bounded common-stage-entry timing diagnostic, not model training or broad recovery scaling. Completed results, supplied cluster exports, proposed protocols, supervisor-conditioned operators, and deployable intervention evidence remain explicitly distinct.

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1 Executive recommendation

1.1 Recommended scientific thesis

The research plan remains viable, but the central contribution should no longer be described as "SAFE plus early triggering plus critic-guided flow." Recent work already studies calibrated VLA verification, recovery windows, counterfactual recovery labels, subtask backtracking, and critic- or reward-guided action generation [3, 4, 5, 6, 7, 8, 9]. The stronger and more defensible thesis is:

> **A failure detector's confidence is not its intervention value. Long-horizon VLA recovery has a state-, stage-, delay-, and operator-dependent deadline that must be measured causally and predicted directly.**

The method should therefore learn whether intervention has positive value, when that value disappears, and which recovery action or operator should be selected. SAFE remains scientifically important as (i) the internal action-policy representation being adapted, (ii) a strong terminal-failure baseline, and (iii) a candidate input to the recovery-value model. It should not be assumed to be a value function.

1.2 Primary paper endpoint

The final primary endpoint is not detector ROC-AUC. It is the equal-task absolute improvement in final RoboCasa365 task success produced by the complete deployable system relative to the strongest frozen non-oracle baseline, under a matched intervention and inference budget. Detector ranking, warning time, stage recovery, and candidate-selection accuracy are mechanistic secondary endpoints that explain the task-success result.

Recommended contribution statement. The paper introduces exact-prefix counterfactual recoverability curves, demonstrates that terminal-failure likelihood and causal intervention value diverge, and learns a stage-aware utility gate that decides whether, when, and how to recover while controlling harmful interventions.

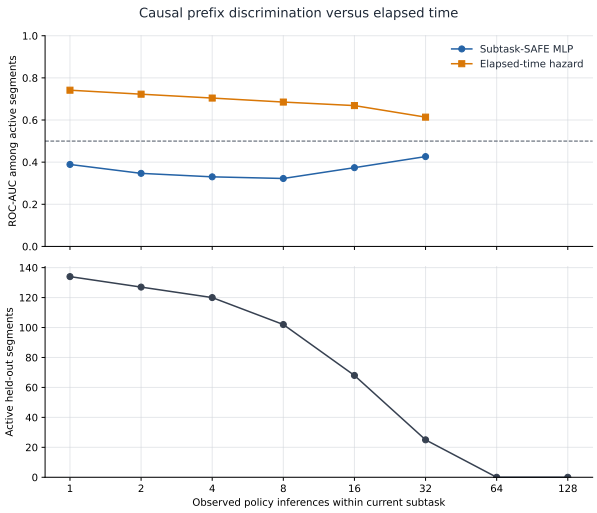
1.3 Starting evidence

Table 1 summarizes the evidence already available in the repository. The strongest current detector evidence is a frozen prospective Xiaomi shadow test, but it did not execute recovery. The five-task stage-aware development gain was tested on 250 new identity-disjoint parents and did not transfer. Phase 2 established exact complete-snapshot branching on a bounded two-task validation. Phase 3 found zero completion across 80 short-suffix candidates. Phase 3B then extended the same exact branches to 30 genuine replans for ArrangeTea and 16 for CuttingToolSelection. Its 140-branch engineering audit passed, but the full screen found only one mixed snapshot, 1.25 percentage points of oracle-over-random headroom, and no full-task successes. This is sufficient to reject local chunk reranking for the frozen two-task candidate set. The subsequent higher-level operator screen completed 160 scientific suffixes with all engineering gates passing, but continuation reached 22.5% stage completion versus 17.5% for every intervention. Oracle operator routing added zero completion gain and no operator passed. The active proposal is now one common-stage-entry timing diagnostic; candidate-critic training, operator-router training, and broad surface collection remain blocked.

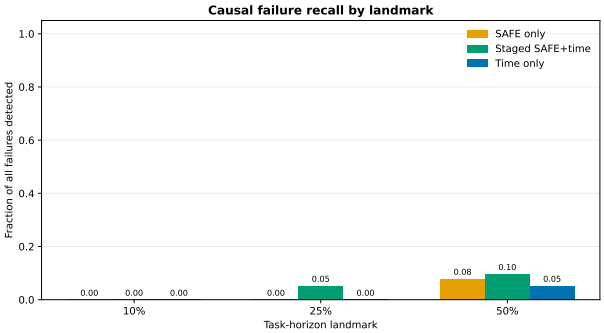
Table 1: Current evidence and the strongest defensible interpretation. "Prospective shadow" means the detector bundle was frozen before scoring new trajectories, but the detector did not change policy execution.

Component	Evidence	Current interpretation
Xiaomi matched-FPR shadow test	760 balanced primary rollouts. Staged SAFE plus time reached TPR 0.668 at FPR 0.050, versus time-only TPR 0.632 at FPR 0.047. The timing gain was 0.0259 task horizons and its bootstrap interval crossed zero; only 2.9% of failures were detected by the 25% landmark.	Modest prospective classification gain; overall early-warning gate failed; no recovery claim.
Full-parent stage-aware confirmation	Development selection favored conditioned SAFE (ROC 0.696 versus 0.592 terminal). On 250 frozen, identity-disjoint parents, conditioned SAFE reached 0.489 versus 0.523 terminal, 0.563 SAFE38, and 0.500 time-only. All preregistered gates failed.	The development gain did not replicate. Predicate-stage-aware SAFE remains a baseline or model input, not a deployable trigger contribution.
State-context follow-up	A new 250-parent cohort recorded a 240-dimensional proprioceptive state history. Development selected stage SAFE at 0.693, context at 0.691, and terminal at 0.690. On the locked 85-parent retrospective holdout, stage reached 0.584, context 0.566, terminal 0.470, and time-only 0.500; both primary contrast intervals crossed zero.	Descriptive evidence of task-specific signal, not prospective confirmation. State context did not improve the selected ranking or the matched-FPR operating point.
RLDX Subtask-SAFE	Raw segment ROC 0.734 fell to parent-aware ROC 0.558, while an elapsed-time baseline reached 0.740. Frozen causal-prefix variants remained near chance.	Retrospective segment separation did not translate to online intervention evidence.
Recovery infrastructure	Ordered subtask anchors, environment reset, EEF recovery, continuation, and Xiaomi deterministic best-of-K proposals exist.	Useful implementation base; existing runs do not establish randomized recovery utility.
Complete snapshot replay	On two tasks, 10 of 10 eligible parents produced 20 of 20 snapshots and 140 of 140 branches: 80 explicit candidates, 40 same-seed repeats, and 20 environment-only controls. There were zero branch errors, exact repeated requests/actions/observations/causal transitions, 100% repeated-suffix agreement, and 100% candidate diversity.	Phase 2 engineering validity passed and supplied the immutable states and branches used by Phases 3 and 3B.
Counterfactual opportunity pilot	The 10-parent, 20-snapshot Phase 3 analysis found zero stage completions and zero full-task successes among 80 candidates. Mixed-outcome fraction, random and oracle completion, oracle headroom, nominal completion, and lowest-SAFE completion were all zero. All candidates ended at the 64-step suffix after only four genuine replans.	The short-suffix proposal–horizon pair had no measurable ranking opportunity. It motivated one bounded training-horizon continuation rather than critic training.

Component	Evidence	Current interpretation
Training-horizon continuation	Phase 3B passed all engineering gates on 10 parents, 20 snapshots, 140 branches, and 140 same-runtime prefix references with zero errors. Among 80 candidates, task-macro random stage completion was 0.2375, oracle-best-of-four was 0.2500, and nominal was 0.2000. Only one snapshot from one ArrangeTea parent was mixed; oracle-minus-random was 0.0125, oracle-minus-nominal was 0.0500, and full-task success was zero.	The local-candidate scientific gate failed. Candidate-critic and five-task candidate scaling are blocked. The subsequently completed operator screen is reported below.
Higher-level operator screen	160 scientific suffixes on the same 10 parents and 20 snapshots; zero errors and every engineering gate passed. Continuation reached 22.5% stage completion and 20.0% endpoint retention; every intervention reached 17.5% completion. No parent benefited on average, oracle routing gain was zero, and full-task success was zero.	Completed negative developmental screen. No useful fixed operator or router target; next test a bounded common-stage-entry timing protocol, not SAFE-triggered recovery.



(a) Causal prefix evaluation exposes the gap between retrospective Subtask-SAFE ranking and online utility.



(b) The frozen Xiaomi shadow test achieved low early-landmark recall on composite failures.

Figure 1: The research plan begins from negative causal and timing evidence rather than from retrospective ROC alone. Figures are reproduced from the checksum-backed full report.

1.4 Dependency-gated roadmap

Table 2: Experiment dependency structure. Each expensive phase is conditional on the preceding scientific or engineering gate.

Phase	Experiment	Positive result	If the gate fails
0	Protocol freeze and evidence audit	Artifacts, identities, and endpoints are immutable and reproducible.	Repair integrity; do not collect.
1	Frozen stage-aware confirmation ✗	Early stage-aware ranking transfers to new parents.	**Observed: gate failed.** SAFE becomes a baseline/input, not the trigger contribution.
2	Complete snapshot replay ✓	**Observed: gate passed.** Same state and randomness reproduced while declared candidate seeds created controlled alternatives.	Not applicable; preserve the passing bundle and hashes.
3	Short-horizon opportunity screen ✗	**Observed: bounded gate failed.** Mixed-outcome fraction and oracle headroom were both zero across 80 candidates.	Preserve the negative result; do not train a critic or reinterpret generic progress as success.
3B	Training-horizon continuation ✗	**Observed: engineering passed, scientific gate failed.** Only 1/20 snapshots was mixed; oracle-minus-random was 0.0125 and no candidate achieved full-task success.	Local chunk reranking is closed for this frozen set; do not train or scale a candidate critic.
3B-R	Higher-level operator screen ✗	**Observed: engineering passed, scientific gate failed.** Continuation was best; every intervention lost five completion points and oracle routing gain was zero.	Preserve the result; no router or fixed executor is supported at these states.
3B-T	Common-stage-entry timing diagnostic ●	Earlier application yields useful paired operator advantage under a shared base and endpoint.	If useful value remains absent, redesign the executor or report a diagnostic benchmark.
4	Recoverability surfaces	Intervention value varies meaningfully with delay, stage, and operator.	Narrow to a diagnostic benchmark or improve operators.
5	Recovery-value critic	The learned critic predicts and ranks causal utility on sealed parents/tasks.	Do not guide the policy; preserve the benchmark result.
6	Prospective best-of-K selection	Critic selection beats random $K = 4$ under identical candidates.	No flow guidance; critic is not actionably causal.
7	Frozen end-to-end intervention	Final task success improves without excessive harm.	No deployable recovery-system claim.

Phase	Experiment	Positive result	If the gate fails
8	Differentiable flow guidance	Guided $K = 1$ matches critic $K = 4$ at lower cost and no safety loss.	Retain discrete critic reranking.
9	Generalization and physical validation	Benefit survives held-out tasks/scenes and another policy or robot.	Scope the paper to same-policy RoboCasa365.

Planning thresholds. The numerical gates in this report are proposed research decisions, not official venue requirements. Final powered sample sizes must be frozen from blinded pilot variance and the exact hierarchical analysis.

Current decision. Preserve Phase 2 and the completed negative candidate and operator screens. The 160-suffix operator screen found no fixed-operator benefit and zero oracle routing headroom; training a router or scaling those arms is unsupported. Run at most one separately registered common-stage-entry timing diagnostic (Section 8.3), followed by fresh-parent validation only if it identifies useful intervention value. SAFE-triggered recovery remains untested and deferred.

2 Formal scientific framework

2.1 Failure risk and recovery value are different estimands

Let h_t denote all causal information available at genuine policy inference t: observations, proprioception, task language, executed action history, current cached-plan state, and any online stage estimate. Let $Y \in \{0, 1\}$ denote eventual full-task success. A conventional failure monitor estimates

$$p_{\text{fail}}(h_t) = \Pr(Y = 0 \mid h_t). \tag{1}$$

This quantity can be high even when no available recovery action improves outcome, and it can be low while a cheap early intervention would be beneficial. The proposed causal object is intervention-specific recoverability. For recovery operator k applied after delay d,

$$R_k(h_t, d) = \Pr(Y = 1 \mid \text{do}(k \text{ at } t + d), h_t). \tag{2}$$

The intervention advantage is

$$\Delta_k(h_t, d) = R_k(h_t, d) - R_{\text{continue}}(h_t, d) - \lambda C_k - \mu H_k, \tag{3}$$

where C_k is compute or latency cost, H_k is intervention harm, and λ, μ are frozen decision weights. A deployable controller intervenes only when the lower-confidence estimate of $\max_k \Delta_k$ exceeds a predeclared threshold, and chooses the operator with the largest predicted value.

2.2 Recovery deadline

The project should not assume that recoverability decays monotonically. A policy may self-correct, enter a more controllable state, or transition between stages. For a positive-value margin ϵ, define the sampled operational deadline

$$D_k(h_t; \epsilon) = \sup\{d \in \mathcal{D} : \Delta_k(h_t, d) > \epsilon\}, \tag{4}$$

where $\mathcal{D}$ is a prespecified delay grid in genuine policy inferences. The scientific question is therefore not simply whether "earlier is better," but how D_k and the shape of $\Delta_k(h_t, d)$ depend on task, stage, failure mode, and operator.

2.3 Counterfactual identification in simulation

For snapshot i, candidate/operator k, delay d, and common-randomness schedule ω, let $Y_i(k, d; \omega)$ be the potential suffix outcome. RoboCasa simulation permits repeated branches from an identical state. This does not identify every real-world causal effect, but it permits controlled estimation of within-simulator intervention contrasts when four conditions hold:

1. the simulator and policy state are fully restored;
2. intervention branches share the same environment and fallback-policy randomness until treatment induces divergence;
3. branch states are selected independently of the detector being evaluated;
4. all branches, including homogeneous candidate sets and harmful interventions, are retained.

Under these conditions, the paired branch contrast

$$Y_i(k, d; \omega) - Y_i(\text{continue}, d; \omega) \tag{5}$$

is the fundamental training and evaluation signal. The top-level population claim still requires resampling tasks and parents; thousands of candidate branches do not create thousands of independent tasks.

2.4 Experimental hierarchy

Table 3: Statistical hierarchy and grouping rules.

Unit	Definition	Required rule
Task	One RoboCasa365 task identity and its semantic-stage catalog.	Top-level resampling unit for cross-task claims.
Parent	One original rollout identified by task, environment seed, reset index, and policy-sampling family.	Split before extracting stages, prefixes, or snapshots.
Snapshot	One complete causal simulator-policy state nested within a parent.	Every operator and candidate branch stays in the same split.
Candidate branch	One action chunk or recovery operator executed after restoration.	Paired observation, not an independent sample.
Inference	One genuine policy model request.	Store one SAFE/context feature; cached actions do not create new samples.
Model seed	One neural initialization.	Measures training instability, not environment or task replication.

2.5 Primary estimands

Frozen detector confirmation. At exact active-stage prefixes 1 and 2, define the task-and-stage macro ROC θ_{rank}, with equal weight for every supported task-stage stratum. The primary contrast is

$$\Delta_{\text{rank}} = \theta_{\text{conditioned}} - \theta_{\text{terminal}}. \tag{6}$$

Candidate actionability. For identical candidate sets,

$$\Delta_K = \Pr(Y = 1 \mid K = 4, \text{critic}) - \Pr(Y = 1 \mid K = 4, \text{random}). \tag{7}$$

End-to-end task success. For method a and frozen control c, the primary deployment estimand is the equal-task treatment effect

$$\Delta_{\text{success}} = \text{mean}_{i \in \mathcal{T}}[\Pr(Y = 1 \mid A = a, i) - \Pr(Y = 1 \mid A = c, i)]. \tag{8}$$

Successful-parent harm. The safety quantity is episode-level: a would-be successful parent that is made unsuccessful by intervention counts as harm. Per-inference false-positive rate is secondary and cannot replace this unit.

2.6 Evidence labels

Every result table and caption should label evidence as one of: *development selection*, *retrospective locked*, *prospective shadow*, *prospective randomized intervention*, *simulator oracle*, or *physical deployment*. Local unit tests and source inspection establish implementation properties only; they are not simulator or policy-performance evidence.

3 Phase 0: Freeze the protocol and audit experimental readiness

Scientific purpose. Establish an immutable, machine-auditable foundation before any new scientific outcome is observed. This phase prevents hidden model selection, identity reuse, path drift, and accidental promotion of a development result into a confirmation claim.

3.1 Scientific rationale

Phase 0 has no performance hypothesis. Its role is to make every later hypothesis falsifiable. A confirmatory result is interpretable only when the policy, feature layer, model, score transformation, task/stage catalog, prefixes, thresholds, seed families, retry rules, and statistical code were fixed before outcomes. Integrity failures are engineering failures, not negative scientific findings.

The current project spans local source code, cluster source, durable artifacts, multiple policy families, and several already opened cohorts. Therefore, the protocol must explicitly distinguish the 170-parent stage-aware development allocation, the opened 80-parent outer set, the prospective Xiaomi shadow cohort, new confirmation identities, critic development identities, calibration identities, and final randomized-intervention identities.

3.2 What must be implemented

1. **Protocol manifest.** Add a machine-readable manifest containing repository commits, policy/checkpoint hashes, SAFE revision, task and stage catalogs, feature layer and aggregation, prefixes, model seeds, normalization, thresholds, candidate count, replan length, recovery horizon, primary endpoint, primary comparison, and bootstrap seed.
2. **Identity registry.** Represent each parent by the tuple

   $$(\text{task}, \text{environment seed}, \text{reset index}, \text{policy-sampling family}).$$

   The registry must reject overlap across development, selection, calibration, confirmation, critic test, and final intervention roles.
3. **Artifact validation.** Verify checksums for the Xiaomi checkpoint, SAFE checkpoints, runtime bundles, server patches, and source commits. Refuse to run if a frozen artifact differs.
4. **Action-equivalence smoke.** Confirm that requesting SAFE/context features with an identical sampling seed does not change the decoded action chunk relative to the action-only request.
5. **Run metadata.** Every run must store exact command, task list, code revision, checkpoint, seed/reset allocation, GPU and port mapping, model-loaded log, output path, and partial/error status.

Existing modules to reuse. Use `docs/cluster_experiment_runbook.md`, `robocasa/recovery/safe/run_stage_aware_parent_safe.py`, `robocasa/recovery/safe/stage_aware_parent_safe.py`, `robocasa/recovery/xiaomi_robotics_1_policy.py`, and the current dataset/schema validators.

Proposed additions. Add `robocasa/recovery/safe/experiment_protocol.py`[new] for schema validation and `robocasa/recovery/safe/audit_experiment_identities.py`[new] for cross-manifest overlap checks. Add unit tests that deliberately alter a checksum, reuse a seed/reset identity, change a frozen prefix, and duplicate one inference record.

3.3 Evaluation

Phase 0 is evaluated with deterministic integrity tests rather than ROC or success rate. The required checks are listed in Table 4.

Table 4: Phase 0 readiness checks.

Check	Evaluation	Positive criterion
Artifact integrity	Recompute hashes and reload every frozen runtime.	Exact hash agreement; strict model loading.
Identity disjointness	Compare all role manifests by rollout ID and task/seed/reset tuple.	Zero overlap.
Feature/action alignment	Repeat seeded action-only and feature-return requests.	Numerically identical decoded action.
Inference sampling	Count genuine server requests and stored feature records.	Exactly one feature per genuine inference.
Cluster preflight	Verify storage, model-loaded log, listening port, server/GPU process, and one bounded rollout.	Zero traceback; complete output and manifest.
Analysis determinism	Re-run the frozen scorer and bootstrap with the same seed.	Identical scores and intervals within numerical tolerance.

Positive result. All frozen artifacts resolve, identity roles are disjoint, the feature-return path is action-equivalent, and one bounded live rollout produces aligned policy, feature, stage, and outcome records. This result establishes experimental readiness only; it does not support a detector or recovery claim.

If the result is negative. Stop new collection. Repair provenance, state alignment, server compatibility, storage, or identity allocation without changing the scientific endpoints. A Phase 0 failure invalidates the affected experiment rather than falsifying a scientific hypothesis.

3.4 Required artifacts

- `protocol.json` and a human-readable protocol summary;
- `allocation_manifest.jsonl` with all planned arms and identities;
- `artifact_checksums.json` and source/checkpoint revisions;
- preflight and one-rollout smoke logs under durable `/gs/bs` storage;
- a PASS/FAIL audit report generated before Phase 1 outcomes are opened.

4 Phase 1: Frozen confirmation of full-parent predicate-stage-aware SAFE

Scientific purpose. Completed; confirmatory gate failed. The promising 0.696 development result was tested on 250 genuinely new parent and seed/reset identities without refitting, renormalization, threshold update, or stage-support change. The result did not transfer.

4.1 Scientific hypothesis

The working explanation for composite-task difficulty is label and temporal-context misalignment. Terminal labels ask whether the entire parent eventually fails. Subtask-SAFE evaluated on independent stage windows discards the parent history and broadcasts an eventual stage outcome across all early samples. The proposed correction preserves the full parent trajectory while conditioning the representation on active stage, stage-anchor feature change, causal progress, task/stage identity, and near-failure horizons. In the current evaluator, however, active-stage labels come from simulator predicates. They therefore provide an oracle-stage estimate of representation quality and must not be confused with a causal online stage estimate.

The confirmatory hypothesis is

$$H_1 : \quad \Delta_{\text{rank}} = \theta_{\text{conditioned}} - \theta_{\text{terminal}} > 0, \tag{9}$$

where θ is the equal-task/equal-stage ROC averaged over exact active-stage prefixes 1 and 2. A positive result would demonstrate early stage-specific ranking signal. It would not yet demonstrate calibrated online warning or recovery utility.

4.2 Completed implementation and frozen protocol

The confirmation path collected a pristine context-enabled cohort and enforced the frozen runtime contract.

1. Fifty fixed-attempt parents were retained for each of the five predeclared composite tasks: 250 parents total, 120 successes and 130 failures. The task names and effects are reported in Table 7.
2. The seed/reset family was disjoint from the 170 development parents and the opened 80-parent outer set. Every valid attempt was retained.
3. Raw Xiaomi SAFE features, decoded action chunks, semantic traces, and causal state history were recorded. The context is a 240-dimensional flattened proprioceptive state history, not RGB or a visual embedding, and it was forbidden as an input to the frozen confirmation.
4. The frozen conditioned, terminal, multihorizon, stage, prototype, time-only, and broad SAFE38 comparators were applied. No model, normalizer, threshold, supported-stage rule, or score transformation was updated on test.
5. The evaluation contained 431 stage events. One failed parent ended in a failed stage with zero genuine policy inferences; it was reported as causally unobserved rather than imputed or silently dropped.

Existing implementation. The primary entry point is `robocasa/recovery/safe/run_stage_aware_parent_safe.py` in `evaluate` mode, with modeling and grouped metrics in `robocasa/recovery/safe/stage_aware_parent_safe.py`. The frozen procedure is documented in `docs/safe_xiaomi_stage_aware_parent.md`.

Tests to add or retain. Tests must verify exact rollout-ID agreement for external SAFE38 scores, rejection of development identity reuse, prohibition of test-time threshold updates, exact prefix aggregation, task/stage macro weighting, and preservation of the context channel without reading it in the confirmation arm.

Deployability boundary. A positive result in this phase authorizes a simulator-predicate-stage-conditioned upper bound only. Phase 5 must replace predicate stage with a frozen causal stage tracker and quantify the oracle-versus-predicted-stage gap before the method can be used online.

4.3 Frozen confirmation result

Table 5: Development selection versus frozen identity-disjoint confirmation. ROC-AUC is the equal-task/equal-stage average over active-stage prefixes 1 and 2. Confidence intervals are hierarchical task/parent bootstrap intervals for the paired contrast.

Quantity	Development	Frozen confirmation	Interpretation
Parents	170 total; 37 selection	250 identity-disjoint	Confirmation identities were never used for fitting or selection.
Conditioned SAFE ROC	0.696	0.489	The development advantage did not transfer.
Terminal SAFE ROC	0.592	0.523	Terminal SAFE exceeded conditioned SAFE on confirmation.
SAFE38 independent ROC	Not primary	0.563	Best frozen SAFE comparator, but still only modest early ranking.
Time-only ROC	0.500	0.500	Exact early prefixes remove elapsed-time ranking by construction.
Conditioned minus terminal	+0.104 descriptive	−0.034; 95% CI [−0.146, +0.076]	Direction reversed and interval crossed zero.
Conditioned minus time	+0.196 descriptive	−0.011; 95% CI [−0.119, +0.112]	No evidence beyond the time control.
Conditioned minus SAFE38	Not frozen	−0.073; 95% CI [−0.243, +0.111]	No advantage over the broad frozen detector.

At the frozen matched-FPR operating point, conditioned SAFE reached failed-stage TPR 0.145 at successful-parent FPR 0.067, compared with time-only TPR 0.504 at FPR 0.033. Miss-adjusted detection fraction was 0.965 for conditioned SAFE and 1.000 for time-only, where a lower fraction is earlier. Thus the detector was neither more sensitive nor operationally earlier than the time baseline.

Table 6: Preregistered Phase 1 gate audit. Every criterion was evaluated without test-time updates.

Criterion	Observed	Verdict
Conditioned macro ROC at least 0.60	0.489	✗
Conditioned minus terminal at least +0.05	−0.034	✗
Hierarchical 95% interval entirely above zero	$[-0.146, +0.076]$	✗
Conditioned beats terminal on at least four of five tasks	2 of 5	✗
Conditioned beats time-only on at least four of five tasks	2 of 5	✗
Matched-FPR TPR no more than 0.05 below time	difference −0.359	✗

If the result is negative. The preregistered Phase 1 result is negative. Treat the 0.696 development ROC as selection noise or cohort-specific structure. Do not refit on the confirmation parents, promote conditioned SAFE as a deployable trigger, or start another broad detector sweep on this cohort. SAFE remains a baseline and a possible input to a later recovery-value model.

4.4 Post-confirmation state-context development study

After the frozen failure was published, the 250-parent context-enabled cohort was used for a new, explicitly developmental study. It was repartitioned into 92 fit, 35 selection, 38 calibration, and 85 locked retrospective-holdout parents. Because this study began after the cohort had already been scored in the frozen confirmation, the 85-parent result is *retrospective locked-holdout evidence*, not a new prospective claim.

Sixty screening fits and 15 refits compared terminal, stage, multihorizon, conditioned, and state-context arms. The selection ROC values were 0.690 terminal, 0.693 stage, 0.685 multihorizon, 0.653 conditioned, and 0.691 context. Stage was selected, but its advantage over context was only 0.002 and over terminal only 0.003, both smaller than the observed seed variation.

On the frozen 85-parent holdout, stage reached prefix-1/2 macro ROC 0.584, context 0.566, terminal 0.470, and time-only 0.500. Stage minus terminal was +0.113 with 95% CI $[-0.075, +0.300]$; stage minus time was +0.084 with 95% CI $[-0.118, +0.267]$. Stage beat terminal on four tasks but time on only three, and its macro ROC remained below the predeclared 0.60 gate. At matched FPR, stage had failed-stage TPR 0.100 at parent FPR 0.049, versus time-only TPR 0.125 at FPR 0.024. Its miss-adjusted timing contrast versus time was −0.014 with 95% CI $[-0.064, 0.000]$. All preregistered operational criteria failed.

Table 7: Retrospective 85-parent prefix-1/2 ROC-AUC by task. These heterogeneous effects are descriptive and not multiplicity-adjusted.

Task	Stage	Context	Conditioned	Multi	Terminal	Time
ArrangeBreadBasket	0.316	0.173	0.279	0.368	0.376	0.500
ArrangeTea	0.625	0.712	0.673	0.567	0.404	0.500
BreadSelection	0.482	0.569	0.565	0.431	0.324	0.500
CuttingToolSelection	0.838	0.709	0.675	0.849	0.613	0.500
GarnishPancake	0.657	0.664	0.743	0.686	0.636	0.500

Table 8: Retrospective prefix-1/2 ROC-AUC by supported semantic stage. Support is the number of evaluable stage events in the 85-parent holdout; sparse rows are descriptive only.

Task and semantic stage	Support	Stage	Context	Terminal	Time
ArrangeBreadBasket: OpenCabinet_1	16	0.250	0.089	0.643	0.500
ArrangeBreadBasket: PickPlaceCabinetToCounter_2_place	7	0.333	0.083	0.042	0.500
ArrangeBreadBasket: PickPlaceCounterToCounter_3_place	15	0.365	0.346	0.442	0.500
ArrangeTea: PickPlaceCabinetToCounter_2_place	17	0.625	0.712	0.404	0.500
BreadSelection: PickPlaceCabinetToCounter_2_place	14	0.488	0.450	0.363	0.500
BreadSelection: PickPlaceCabinetToCounter_2_release	5	0.625	0.875	0.125	0.500
BreadSelection: PickPlaceCounterToCounter_1_place	17	0.333	0.383	0.483	0.500
CuttingToolSelection: OpenDrawer_1	17	0.697	0.689	0.538	0.500
CuttingToolSelection: PickPlaceDrawerToCounter_2_place	11	0.979	0.729	0.688	0.500
GarnishPancake: strawberry_on_pancake	17	0.657	0.664	0.636	0.500

The task audit identifies a real scientific diagnostic rather than a general detector win. Stage-aware scoring was strong for `CuttingToolSelection` and positive for `ArrangeTea` and `GarnishPancake`, but failed sharply for `ArrangeBreadBasket`. Within `ArrangeBreadBasket`, stage ROC ranged from 0.250 for `OpenCabinet_1` to 0.365 for the third placement stage; state context fell to 0.089 on `OpenCabinet_1`. Conversely, `CuttingToolSelection::PickPlaceDrawerToCounter_2_place` reached stage ROC 0.979. A sparse five-event BreadSelection release stage produced context ROC 0.875 and must not be treated as stable evidence. This heterogeneity argues for measuring intervention value by state and operator, not for adding more global stage-label complexity.

Interpretation of state context. The recorded context is proprioceptive state history only. It did not beat the selected stage arm on the locked holdout and does not establish a visual-context contribution. Any future visual ablation requires causal RGB embeddings and a new untouched cohort.

4.5 Evaluation design

Table 9: Frozen stage-aware confirmation design.

Component	Frozen definition
Primary sampling unit	Parent rollout; inferences and stages remain clustered within parent.
Primary prefixes	First one and two genuine policy inferences of the active stage.
Primary endpoint	Equal-task/equal-stage macro ROC averaged over prefixes 1 and 2.
Primary contrast	Conditioned stage-aware SAFE minus frozen terminal SAFE.
Mandatory controls	Terminal SAFE, stage-conditioned elapsed time, broad SAFE38, and prototype distance.
Uncertainty	Hierarchical bootstrap: resample tasks, then parents, and retain all stages/inferences of each parent; at least 2,000 frozen-seed replicates.
Secondary analyses	Per-task/per-stage ROC and AP; prefixes 4 and 8 with risk-set support; calibration and event metrics labeled descriptive.
Forbidden updates	Model, regularization, epoch, normalizer, stage catalog, prototype, threshold, task support, prefix, or score transformation.

The existing 50-parent-per-task plan is a screening minimum. If the fixed-attempt cohort lacks sufficient attempted-stage failures, the extension rule must be specified before model scores are inspected. For example, continue fixed attempts for all tasks until a predeclared parent cap or class-support target is reached; never extend only favorable tasks.

4.6 What counts as a positive result

All gates should pass:

1. conditioned macro ROC is at least 0.60;
2. conditioned minus terminal is at least +0.05;
3. the hierarchical 95% interval for the contrast is entirely above zero;
4. conditioned beats terminal on at least four of five tasks;
5. conditioned beats time-only on at least four of five tasks;
6. the result is not explained only by `ArrangeTea` or `GarnishPancake`; `CuttingToolSelection` and `ArrangeBreadBasket` remain explicit diagnostics.

Positive result. A passing result means that complete-parent, predicate-stage-conditioned action features contain replicable early ranking information beyond terminal SAFE and task/stage timing on these five tasks. It licenses development of a recovery-value model using SAFE/state-context features and a deployable stage tracker. It does not prove a useful 5% alarm threshold, a recovery deadline, task-success improvement, or deployability without simulator predicates.

4.7 Required outputs

The completed artifacts include the development-versus-confirmation comparison, all five task effects, supported-stage audit, parent/task bootstrap intervals, explicit PASS/FAIL gate table, frozen runtime bundles, and the zero-inference censoring record. The original 18-success calibration and the later 38-parent context calibration remain too small for a precise operational 5% FPR claim; zero empirical calibration FPR is not evidence of exact control.

5 Phase 2: Validate complete simulator-policy snapshot replay

Scientific purpose. Establish that alternative recovery branches differ because of the declared intervention rather than unrecorded simulator state, cached policy actions, observation queues, instructions, counters, or random-number state.

5.1 Scientific hypothesis

The scientific hypothesis is an experimental-identification condition:

> Given an identical complete snapshot and common-randomness schedule, the nominal suffix is reproducible; changing only the declared candidate seed or recovery operator produces an aligned counterfactual branch.

The historical `_capture_state()` path in `robocasa/recovery/recovery_rollout.py` captured environment/simulator state and selected qpos/qvel arrays, but did not by itself guarantee restoration of policy image queues, proprioception queues, cached action chunks, instruction, inference indices, candidate-selection indices, stage-tracker state, or remote sampling state. Phase 2 implemented and tested the stronger complete-state contract needed to attribute branch differences to the declared intervention.

5.2 Implemented snapshot contract

The versioned `FullSnapshot` contains:

- serialized RoboCasa environment state, MuJoCo qpos/qvel, model state where required, task randomization, and environment RNG;
- the exact current observation and ordered subtask trace;
- Xiaomi image queues, state queues, cached action plan, last instruction, environment/inference counters, ordinary-inference index, and candidate-selection index;
- explicit policy/server sampling seed schedule and model request metadata;
- causal stage-tracker distribution, transition state, and completed-stage prefix;
- parent, snapshot, schema, code, and artifact hashes.

The branch runner should restore the complete snapshot, validate its hash, request a declared candidate, execute the first chunk, and continue with a frozen fallback-policy seed schedule. It must log where common randomness ends after treatment-induced state divergence.

Implemented modules. `robocasa/recovery/full_snapshot.py` serializes and validates complete simulator, policy, controller, tracker, observation, and RNG state. `robocasa/recovery/counterfactual_branch.py` restores and executes declared branches. `robocasa/recovery/xiaomi_robotics_1_policy.py` exposes explicit policy-state capture/restoration and a shared restored server connection, avoiding hidden reconstruction of private fields.

Completed validation. Unit tests cover snapshot round trips, controller and RNG state, observation/request/action sequence alignment, branch composition, resume semantics, and combined audits. Live debugging separated noncausal diagnostic sensor caches from causal controller state, restored deterministic environment construction, and enforced a single shared restored policy connection. The final bounded run compared complete request, action, observation, simulator, controller, ordered-progress, and suffix records.

5.3 Bounded validation experiment

Use two diagnostic tasks: `ArrangeTea` and `CuttingToolSelection`. Collect five new parent identities per task and two snapshots per parent: active-stage prefix 2 and the first inference at or beyond a fixed 25% training-derived stage horizon. For each of 20 snapshots, execute:

- two repeated nominal branches with identical seeds;
- four branches using explicit, distinct Xiaomi candidate seeds.

This produces 120 primary policy branches. One environment-only negative control per snapshot adds 20 diagnostic branches, for 140 total.

Table 10: Snapshot-replay validation metrics.

Metric	Scientific role	Proposed positive criterion
Request/action equivalence	Verifies identical policy input and sampling.	Exact seed/request match and numerically identical action.
Immediate state deviation	Detects incomplete simulator restoration.	qpos/qvel within frozen tolerance; identical ordered progress.
Suffix outcome agreement	Tests practical reproducibility.	At least 95% agreement for same-seed repeated branches, with no task-specific failure pattern.
Candidate action diversity	Determines whether the proposal distribution offers alternatives.	Non-identical chunks in at least 90% of snapshots.
Record alignment	Prevents candidate/feature/outcome mismatch.	One-to-one seed, feature, action, and outcome mapping.
Restore-induced regression	Detects predicate or observation corruption.	Zero previously completed-stage regressions caused solely by restoration.
Branch error rate	Measures infrastructure fitness.	Below 2% with all errors retained.

The seed, serialized request, returned action chunk, and first restored transition are exact reproducibility requirements, subject only to a predeclared numerical tolerance for floating-point simulator arrays. The 95% suffix criterion is a secondary practical diagnostic for residual nondeterminism; it is not permission to accept mismatched requests or first actions.

An environment-only reset branch should be retained as a diagnostic negative control. If it differs while the full snapshot reproduces, the experiment directly demonstrates why policy-state restoration

is necessary.

5.4 Completed evidence

The final bounded run completed on 25 August 2026 using the shared-restored policy protocol. Ten eligible parents—five per task—produced 20 complete snapshots and all 140 planned branches. Two additional parent attempts were retained as ineligible because they did not reach both predeclared boundaries; they are reachability exclusions, not missing branch outcomes. The immutable artifact root, complete path, and provenance hashes are retained in the durable Phase 2 manifests.

Table 11: Completed Phase 2 bounded validation. Every predeclared engineering gate passed.

Audit item	Observed result
Parent/snapshot support	10/10 eligible parents and 20/20 snapshots; five parents per task.
Branch support	140/140 total: 80 candidates, 40 same-seed repeats, and 20 environment-only controls; 120/120 primary branches.
Errors and artifacts	Zero errors, unique branch identifiers, no missing payloads, and 20 referenced snapshots.
Exact reproducibility	Requests, request sequences, first actions, action sequences, observations, first transitions, and causal environment sequences all exact.
Practical reproducibility	Same-seed suffix outcome agreement 1.000; no non-reproducing repeat pairs.
Proposal diversity	Candidate-diverse snapshot fraction 1.000.
Restore safety	Zero restore-induced regressions; branch error rate 0.000.

Positive result. Observed. Exact-state branch contrasts are scientifically interpretable within this bounded simulator protocol, and Xiaomi's proposal distribution produced non-degenerate first actions. This does not show that the candidates differ in recovery utility, that SAFE can rank them, or that recovery improves full-task success.

If the result is negative. This gate did not fail in the final bounded run. If a later task or platform violates it, stop that counterfactual collection block and diagnose simulator state, policy cache, observation refresh, instruction, counters, or server sampling rather than averaging irreproducible branches.

5.5 Required outputs

The run retains the snapshot schema, per-snapshot payloads and checksums, branch records, exact-repeat diagnostics, candidate-diversity audit, error ledger, frozen plan, and analysis. Videos remain a human-facing diagnostic; numerical state and trace checks are the validity evidence.

6 Phase 3: Screen for counterfactual recovery opportunity

Scientific purpose. Completed bounded pilot; continuation gate failed. Exact Phase 2 snapshots and candidate branches were analyzed without new simulator calls. Across 10 parents, 20 snapshots, and 80 candidates, no candidate completed its frozen trigger stage or full task. The result rejects the 64-step proposal–horizon pair. It does not evaluate SAFE ranking quality because there was no successful candidate to rank.

6.1 Scientific hypothesis

Even a perfect critic cannot improve recovery if all candidate chunks from a snapshot lead to the same outcome. Before training, estimate the oracle headroom of the proposal set. For candidate set $\mathcal{A}_i = \{a_{i1}, \ldots, a_{iK}\}$ at snapshot i, define

$$O_i = \max_{a \in \mathcal{A}_i} Y_i(a) - \text{mean}_{a \in \mathcal{A}_i} Y_i(a). \tag{10}$$

$O_i > 0$ means candidate selection can outperform random choice for that snapshot. The phase hypothesis is that a nontrivial fraction of early and mid-stage snapshots have mixed candidate outcomes and positive oracle headroom.

6.2 Implemented bounded analyzer

Phase 2 already collected the first bounded candidate set with exact parent, snapshot, candidate, seed, action, raw SAFE feature, state context, suffix outcome, ordered progress, regression, and source provenance. Phase 3 analyzed those existing artifacts without new simulator calls. Each derived candidate row retains an immutable source-payload reference plus compressed-file, payload, action, feature, snapshot, plan, runtime, code, and checkpoint hashes.

Selection policies are computed offline after all candidate outcomes are available. This permits exactly paired comparisons among default $K = 1$, expected random $K = 4$, lowest terminal-SAFE score, highest terminal-SAFE score, action-norm heuristics, and oracle best-of-K without generating different candidate sets.

Existing implementation. Reuse explicit sampling seeds and diversity logging in `robocasa/recovery/xiaomi_robotics_1_policy.py`, frozen SAFE scoring in `robocasa/recovery/safe/xr1_best_of_k.py`, ordered completion in `robocasa/recovery/subtask_eval.py`, exact branches in `robocasa/recovery/counterfactual_branch.py`, and the complete Phase 2 runner in `robocasa/recovery/run_phase2_snapshot_replay.py`.

Current limitation. The completed pilot covers two tasks, 20 snapshots, four explicit candidates per snapshot, and one frozen short suffix. It screens the specific proposal–horizon combination only; it cannot estimate the formal five-task task-support gate, deployment prevalence, long-delay recoverability, or final-task benefit.

Phase 3 analyzer. `robocasa/recovery/analyze_phase3_candidate_opportunity.py` is a read-only paired analyzer. It requires a zero-error all-pass shared-restored Phase 2 source, collapses exactly agreeing nominal repeats, excludes environment-only controls, scores the four explicit candidates, computes task-macro and task–parent–snapshot bootstrap estimates, optionally applies the frozen three-seed independent SAFE ensemble, refuses output overwrite, and verifies that source hashes remain unchanged. Its focused tests include a mixed-outcome opportunity case, invalid Phase 2 rejection, and nominal-repeat disagreement rejection.

6.3 Completed Tier 1 result

The bounded pilot used the all-pass shared-policy Phase 2 artifact: 10 parents, five per task; 20 exact snapshots; and four explicitly seeded candidates per snapshot. The source contained no branch errors, reproduced same-seed causal trajectories exactly, and preserved candidate diversity. The analysis artifact is `xr1_phase3_candidate_opportunity_v3_indep_2tasks_20260826_232556`. SAFE scoring used the frozen independent ensemble and did not update a checkpoint, normalization, threshold, candidate set, or outcome.

Table 12: Observed Phase 3 bounded candidate-opportunity result. All completion quantities are task-macro estimates; both tasks had the same zero-completion result.

Quantity	Observed value
Parents / snapshots / explicit candidates	10/20/80
Mixed-outcome snapshot fraction	0.000
Expected random-of-four completion	0.000
Oracle-best-of-four completion	0.000
Oracle headroom over random	+0.000
Nominal continuation completion	0.000
Lowest-SAFE completion	0.000
Lowest-SAFE minus random	+0.000
Snapshots with nonzero SAFE-score spread	20/20
Formal five-task gate	Not estimable in this two-task pilot
Scale decision	**Redesign; do not train a critic**

The zero opportunity was not caused by identical proposals. Mean pairwise first-action distance was 1.486 for ArrangeTea and 0.520 for CuttingToolSelection; mean SAFE-feature distance was 469.147 and 456.471, respectively. Every snapshot had nonzero SAFE-score spread, although the mean spread was only 1.14×10^{-4} and the maximum was 9.42×10^{-4}. Thus the proposals were numerically distinct, but none produced the required semantic outcome within the frozen suffix.

Table 13: Mechanistic diagnostic of the zero-opportunity result. A policy-inference window is one genuine Xiaomi replan, not one cached environment action.

Diagnostic	Observed value
Frozen suffix	64 environment steps
Xiaomi replan interval	16 environment steps
Policy-inference windows per branch	4
Training-derived stage horizon	ArrangeTea 30 replans; CuttingToolSelection 16 replans
Branches ending at suffix horizon	80/80
Stage completions / full-task successes	0/80 / 0/80
Positive / zero / negative progress	2/80 / 68/80 / 10/80
Mean / maximum progress gain	-0.025 / $+0.25$
Regressions	12/80

The 64-step suffix exposed only four replans, or approximately 13% of the ArrangeTea stage horizon and 25% of the CuttingToolSelection stage horizon. This is the most plausible mechanistic explanation for the absence of completions, but it is not evidence that a longer suffix will succeed. The negative result is instead precise: *the current snapshot, proposal, and 64-step continuation protocol contains no measurable candidate-selection opportunity.*

What was and was not learned. The pilot rejects immediate scaling of this short-suffix protocol and blocks critic training. It does not show that SAFE cannot rank recovery candidates: oracle, random, nominal, and SAFE-selected policies all score zero because no successful candidate exists. It also does not establish that candidate sampling is intrinsically poor, because the frozen horizon was substantially shorter than both training-derived stage horizons.

6.4 Two-tier experimental design

Tier 1: completed bounded pilot. The 20 validated Phase 2 snapshots and 80 explicit candidates were analyzed. The two same-seed repeats collapsed to one nominal control; the environment-only branch remained a replay diagnostic and was excluded from utility estimates. No checkpoint, normalization, threshold, candidate set, or outcome was updated. The frozen live trigger predicate had to be false at the snapshot and become true within the suffix, or the full task had to succeed; generic progress and branch-local ordered traces did not count as completion. The pilot failed both estimable continuation criteria and could not adjudicate the formal "three of five tasks" requirement.

The pilot supported scaling only if at least 20% of snapshots had mixed candidate outcomes and task-macro oracle-best-of-four exceeded expected random-of-four completion by at least 15 percentage points. Both observed values were zero. This requires redesign before collection, not critic training.

Tier 2: blocked conditional formal screen. Do not run the five-task design unless the bounded horizon redesign below first establishes nonzero opportunity. If that redesign passes, use the same five development tasks and the exact Phase 2 snapshot protocol. The larger design below remains frozen before new collection.

Use the same five development tasks. For each task, collect 10 eventually failed and 10 eventually successful-control parents, producing 100 parents. This is an explicit case-control opportunity study and must never be used to estimate deployment prevalence or event rates.

From each parent, branch two states selected independently of SAFE score:

1. exact active-stage prefix 2;

2. the first inference at or beyond 25% of a successful-stage horizon derived only from training data.

At each of 200 snapshots, evaluate the nominal candidate and four explicitly seeded candidates. Execute the selected first chunk and then continue with one frozen fallback-policy seed schedule. The planning total is 1,000 suffixes.

The primary outcome is completion of the frozen live trigger predicate within a frozen horizon, for example a training-derived 90th percentile of successful remaining-stage duration with lower and upper caps. The predicate must be false at the snapshot and become true during the suffix, or the full task must succeed. Branch-local ordered traces restart without the parent's monotonic completed-prefix history and are diagnostic only; they must not redefine the active trigger. Full-task success, progress, regression, safety, and cost are secondary. A candidate that finishes the immediate stage but systematically destroys later task success must not be considered a successful recovery method.

6.5 Evaluation

Table 14: Counterfactual opportunity estimands.

Estimand	Interpretation
Mixed-outcome fraction	Fraction of snapshots containing at least one successful and one unsuccessful candidate.
Oracle headroom	Oracle-best-of-four recovery minus expected random-of-four recovery.
Nominal proposal benefit	Random-of-four minus default $K = 1$, separating sampling benefit from scoring benefit.
SAFE actionability	Lowest-SAFE minus random and lowest-SAFE minus highest-SAFE on identical candidate sets.
Successful-control harm	Probability that intervention breaks a snapshot whose nominal continuation succeeds.
Candidate support	Pairwise action distance, feature distance, numerical score spread, and tie rate.

Use snapshot-level paired contrasts and bootstrap task, then parent, then snapshot while retaining all candidate branches. Equalize snapshot weight so a task with more valid candidate records does not dominate.

Positive result. For the formal Tier 2 screen, proceed when at least 20% of snapshots have mixed outcomes, mixed outcomes occur in at least three of five tasks, and oracle-best-of-four improves recovery by at least 15 percentage points over random choice. Tier 1 uses only the two estimable numerical criteria as a scale decision. A positive result means the action proposal contains exploitable causal recovery variation; it does not mean SAFE or a learned critic can identify the good candidate.

If the result is negative. Observed: do not train a critic and do not scale to five tasks. The distinct Phase 3B training-horizon continuation was completed and still found negligible local-candidate benefit: oracle-minus-random was 0.0125, only one of 20 snapshots was mixed, and full-task success was zero. Local chunk reranking is now closed for this proposal family; the subsequent higher-level operator screen also failed, and the next proposed diagnostic tests intervention timing from a shared stage-entry state (Section 8.3).

6.6 Required outputs

Tier 1 produced immutable candidate references, snapshot/task tables, hashes, estimates, uncertainty, harm, and gates. Phase 3B referenced those immutable artifacts and completed the longer-horizon screen. Tier 2 was not unlocked; no five-task candidate corpus should be collected for the current proposal family.

7 Phase 3B: Extend exact branches to the training horizon and select the recovery mechanism

Scientific purpose. Completed developmental result. Exact long-horizon continuation passed every engineering gate on all 20 Phase 2 snapshots, but the candidate set showed negligible oracle selection headroom and no full-task success. The frozen routing decision was to test higher-level recovery operators; that follow-up also completed negatively and is reported in Section 8.

7.1 Evidence boundary and scientific question

Phase 1, Phase 2, and Phase 3 are immutable completed results. Phase 3B was not a retrospective rescue of the failed 64-step gate: it changed the suffix horizon, used a new manifest and output root, and remains developmental evidence. The scientific question was whether the same snapshots, candidate seeds, policy, trigger definitions, nominal controls, and fallback schedules contained useful causal recovery variation when each branch received enough genuine policy replans to complete its active stage.

Let $Y_{ij} \in \{0,1\}$ indicate whether explicit candidate $j \in \{1,\dots,4\}$ completes snapshot i's frozen live trigger predicate within the task-specific horizon, with full-task success also counted as completion. Let Y_i^{nominal} be the agreeing same-seed nominal continuation. Define

$$M_i = \mathbf{1}\left\{\min_j Y_{ij} < \max_j Y_{ij}\right\}, \tag{11}$$

$$R_i^{\text{random}} = \frac{1}{4}\sum_{j=1}^{4} Y_{ij}, \qquad R_i^{\text{oracle}} = \max_j Y_{ij}, \tag{12}$$

$$O_i = R_i^{\text{oracle}} - R_i^{\text{random}}, \qquad G_i = R_i^{\text{oracle}} - Y_i^{\text{nominal}}, \tag{13}$$

$$B_i = R_i^{\text{random}} - Y_i^{\text{nominal}}. \tag{14}$$

M_i measures rankable variation, O_i is finite-set ranking headroom over random selection, G_i tests whether even the oracle candidate improves over simply continuing, and B_i separates generic proposal or replanning benefit from ranking benefit. Oracle-best-of-four is an optimistic finite-seed upper bound, not a deployable success estimate.

7.2 Phase 3B-0: four-snapshot engineering sentinel

Before inspecting any extended outcome, deterministically choose the lowest-reset-index valid parent from each of ArrangeTea and CuttingToolSelection, retaining its prefix and landmark snapshots. This gives four snapshots. For each snapshot, execute the same four explicit candidate seeds, two same-seed nominal repeats, and one environment-only control, for $4 \times 7 = 28$ branches.

Use 480 environment steps for ArrangeTea (30 genuine replans) and 256 for CuttingToolSelection (16 genuine replans). The sentinel passes only if:

1. the frozen Phase 2 source-plan, payload, snapshot, runtime, code, and checkpoint hashes match;
2. the deterministic snapshot selection and exact recorded candidate and continuation seeds match the registered manifest;
3. every extended branch reproduces its corresponding Phase 2 first 64 requests, actions, observations, causal environment transitions, and semantic traces exactly;
4. same-seed nominal repeats agree exactly, every requested artifact is complete, and there are zero branch errors;

5. each task receives its registered horizon and the runner stops only at stage completion, full-task success, terminal failure, safety termination, or that horizon.

Engineering-only gate. If these integrity conditions pass, run Phase 3B-1 regardless of the four observed recovery outcomes. The sentinel is not a scientific opportunity screen. If the true mixed-outcome probability were 0.20, observing zero mixed snapshots out of four would still have probability $0.8^4 = 0.4096$; the one-sided 95% binomial upper bound after 0/4 is approximately 0.53. A zero result here cannot justify abandoning local reranking.

An integrity failure stops collection and requires repair followed by a new output root; it does not count as negative recovery evidence. The failed sentinel and error ledger remain preserved.

Positive result. Observed engineering result: the corrected sentinel completed two parents, four snapshots, 28 branches, and 28 same-runtime prefix references. Every registered first-64 channel was exact, all restore and provenance gates passed, and there were zero branch errors. This unlocked the full screen; the sentinel outcomes were not used as an efficacy stop.

7.3 Phase 3B-1: all 20 existing snapshots

After the sentinel passes, extend all 20 Phase 2 snapshots with the identical seven-branch design, producing 140 long-horizon branches. Do not select snapshots based on the sentinel outcomes. Scientific estimates use the four explicit candidates; nominal repeats are collapsed only after exact-agreement validation, and environment-only branches remain replay controls rather than recovery arms.

The snapshot is the primary paired unit, nested within parent and task. Report task-macro estimates and preserve all branches from a sampled snapshot during uncertainty analysis. With only two tasks, task signs are guardrails and intervals are cohort-specific; they cannot establish generalization to the RoboCasa365 task population.

Primary and durability outcomes. Report completion of the frozen live predicate at every genuine replan, the risk set remaining eligible at each replan, random-of-four minus nominal, oracle-minus-random, and oracle-minus-nominal. Separately report full-task success when observed, prior-stage regression, safety termination, generic ordered progress, compute, and whether an immediate stage completion remains valid through the next frozen stage landmark or a common terminal horizon. Generic progress must not replace the binary primary outcome.

The current Phase 2 parents were not all continued to terminal completion. Consequently Phase 3B cannot estimate natural deployment prevalence or eventual-success-control harm. It can measure branch-local regression, safety, and durability relative to the same snapshot; a formal successful-control analysis requires terminal parent labels.

7.4 Observed full-screen result

The full run completed all 10 parents, 20 snapshots, 140 long-horizon branches, and 140 same-runtime prefix references with zero errors. Every engineering gate passed. The scientific analysis was therefore estimable, but all six candidate-opportunity gates failed. Table 15 reports the frozen task-macro and per-task estimates.

Table 15: Observed Phase 3B training-horizon outcomes. Values are fractions over the registered snapshot/candidate support.

Metric	Task macro	ArrangeTea	CuttingToolSelection
Mixed-outcome fraction	0.0500	0.1000	0.0000
Random-of-four stage completion	0.2375	0.4750	0.0000
Oracle-best-of-four stage completion	0.2500	0.5000	0.0000
Nominal stage completion	0.2000	0.4000	0.0000
Oracle minus random	0.0125	0.0250	0.0000
Oracle minus nominal	0.0500	0.1000	0.0000
Durable completion fraction	0.2500	0.5000	0.0000
Candidate regression rate	0.5875	0.8750	0.3000
Candidate full-task success	0.0000	0.0000	0.0000
Safety termination rate	0.0000	0.0000	0.0000

Only one of 20 snapshots, belonging to one ArrangeTea parent, had mixed candidate outcomes; CuttingToolSelection had none. The paired hierarchical bootstrap placed oracle-minus-random at 0.0128 with a 95% interval of $[0, 0.05]$, oracle-minus-nominal at 0.0516 with $[0, 0.20]$, and the mixed fraction at 0.0512 with $[0, 0.20]$. ArrangeTea completion remained zero through 256 environment steps and reached 0.475 by 448–480 steps, while CuttingToolSelection remained at zero through its 256-step horizon. Thus the longer horizon revealed delayed stage completion in one task but almost no opportunity to choose among four stochastic local chunks.

The absolute candidate regression rate was high, especially for ArrangeTea, although it was 0.0625 lower than nominal in the task-macro comparison and therefore passed the registered non-increase guardrail. There were no catastrophic safety terminations, and observed stage completions were durable under the registered criterion. These safety and durability passes do not rescue the failed opportunity gate: even oracle candidate selection improved random choice by only 1.25 percentage points and no branch achieved full-task success.

7.5 Developmental continuation gates

Local candidate-critic work becomes eligible only if the full 20-snapshot screen satisfies all of the following planning thresholds:

1. at least 4/20 snapshots have mixed candidate outcomes, arising from at least three independent parents and both tasks;
2. task-macro oracle-over-random headroom is at least 15 percentage points;
3. task-macro oracle-over-nominal headroom is positive in both tasks and approximately at least 10 percentage points overall;
4. there is no catastrophic safety violation and the task-macro increase in prior-stage regression or declared safety harm is at most five percentage points relative to nominal;
5. apparent immediate completions retain their benefit at the next stage landmark or common terminal horizon.

These are compute-allocation thresholds, not venue requirements or confirmatory tests. If 0/20 snapshots are mixed, the simple independent-binomial one-sided 95% upper bound is approximately 0.14 and $P(0/20 \mid p = 0.20) \approx 0.0115$. Because the 20 snapshots are nested within 10 parents and only two tasks, report those calculations as descriptive sensitivity, not population confidence.

If the result is negative. Observed: mixed support occurred in only one snapshot and one parent, in ArrangeTea only; oracle-minus-random was 0.0125 rather than 0.15; oracle-minus-nominal was zero for CuttingToolSelection and 0.05 task-macro rather than 0.10; and full-task success was zero. The formal gate was estimable and failed. Do not train a candidate critic, increase K, or scale this local proposal family to five tasks. The registered routing decision is `higher_level_operator_screen`.

Table 16: Mechanism selection after the full Phase 3B-1 screen.

Observed pattern	Scientific decision
Mixed outcomes, with oracle beating both random and nominal	A candidate-conditioned critic is plausible; proceed to the formal five-task candidate-opportunity screen.
Candidates are mostly homogeneous successes and random candidates beat nominal	Fresh proposal or replanning helps, but ranking is unnecessary; test a simple frozen fresh-replan policy.
Mixed outcomes exist, but oracle does not beat nominal	The set may be rankable relative to random but is operationally inferior to continuation; do not train a deployment critic.
Completion or mixed support remains rare (observed)	**Stop local chunk reranking. The higher-level follow-up below subsequently completed negatively.**

7.6 Fallback: bounded higher-level operator-headroom screen

The local-candidate gate selected this follow-up, which has now completed. The executed design compared `continue`, `history_reset_replan`, `stage_retry`, and `retreat_reset_replan` on the same 20 snapshots, with two paired continuation-seed schedules, for 160 scientific suffixes. Cache-only flush was replaced by history reset because the inference-boundary snapshots had an empty cache. Stage retry used fixed supervisor-supplied goals. The earlier planned 20 oracle-rewind branches were omitted because genuine earlier complete anchors were unavailable; no 180-branch or rewind result is claimed.

All engineering gates passed, but continuation reached 22.5% stage completion versus 17.5% for every intervention, oracle operator routing added zero completion gain, and full-task success was zero. The completed result and practical-gate failure are detailed in Section 8. The resulting decision is `mechanism_redesign_or_earlier_snapshot_study`, not router training or broad recoverability-surface collection.

At most one separately registered common-stage-entry timing diagnostic is proposed in Section 8.3. It must use real earlier simulator-policy states, paired nominal prefixes, and a common absolute endpoint; it cannot relabel current snapshots or claim that the failed screen established a timing mechanism.

7.7 Conditional formal five-task screen

Only the first row of Table 16 unlocks the candidate-ranking screen, and the observed result did not enter that row. The formal five-task candidate screen is therefore blocked for the current proposal family. If a materially different future proposal mechanism first establishes local candidate headroom under a separately frozen pilot, the conditional design would use five tasks, 10 eventually failed and 10 eventually successful-control parents per task, and two detector-independent snapshots per parent, producing 200 snapshots and approximately 1,000 suffixes. This would remain a case-control opportunity study, not an estimate of failure prevalence.

The formal gate retains at least 20% mixed snapshots, support in at least three of five tasks, at least 15 points of task-macro oracle-over-random headroom, positive oracle-over-nominal benefit, next-

stage durability, and bounded harm. The collector must continue each nominal parent to terminal outcome before extracting the two snapshots; the current Phase 2 runner stops after its snapshot pair and therefore cannot provide the eventual failed/successful-control labels required by this design.

7.8 Implementation and tests

The existing Phase 2 run cannot be legally resumed with a longer suffix because its frozen plan records one global suffix length and resume validation rejects a change. Implement a separate immutable runner and analyzer rather than weakening that validator.

Implemented modules. Phase 3B executed through `robocasa/recovery/run_phase3b_training_horizon_continuation.py` and `robocasa/recovery/analyze_phase3b_training_horizon.py`, reusing `robocasa/recovery/full_snapshot.py`, `robocasa/recovery/counterfactual_branch.py`, per-branch `BranchSpec` horizons, stored per-step hashes and traces, and exact seed records. The immutable runner used a separate output root and resume ledger and failed closed during development on prefix, checkpoint, XML, and provenance mismatches before the final valid run.

Focused tests must cover deterministic snapshot selection, frozen source hashes, task-specific horizons, exact recorded seed reuse, equality of every first-64 channel, deliberate mismatch failure, nominal-repeat agreement, error preservation, resume without overwrite, and every scientific gate and routing case.

7.9 Required outputs

The completed sentinel and full-screen registrations, deterministic selection manifest, source and output hashes, branch-completeness ledger, first-64 equality table, genuine-replan completion curves, task/parent/snapshot support, M_i, O_i, G_i, and B_i summaries, durability and harm results, and frozen routing decision are preserved in the durable Phase 3B artifact root:

```
/gs/bs/tga-shinoda/felid/robocasa_checkpoints/safe/
xr1_phase3b_terminal_camera_v7_full20_20261012_091253.
```

Representative homogeneous, mixed, beneficial, regressive, and oracle-only videos should be selected by a frozen rule rather than after method comparison.

8 Phase 3B-R: Completed higher-level operator screen and next timing diagnostic

Scientific purpose. Completed negative developmental result. All 160 scientific suffixes passed the engineering audit, but continuation outperformed every intervention and finite-seed oracle routing added no completion gain. The next proposed experiment is a small common-stage-entry timing study; model training and broad recovery scaling remain blocked.

8.1 Executed design and provenance

The screen reused the 10 parents, 20 complete snapshots, and two tasks from Phase 2. It executed four operators under two paired continuation-seed schedules: $20 \times 4 \times 2 = 160$ scientific suffixes, following an engineering-only sentinel. The full audit reports 160 unique branches, zero errors, and passing snapshot, restore, request-alignment, seed-pairing, budget, sentinel-repeat, history-reset, and retreat-controller gates. Sentinel outcomes did not determine whether the full screen ran.

The executed interventions were `continue`, `history_reset_replan`, `stage_retry`, and `retreat_reset_replan`. History reset replaced the originally proposed cache-only flush because the snapshots were already at inference boundaries with an empty cache. Stage retry used a fixed *supervisor-supplied* task-specific goal for 64 steps; it did not validate a deployable stage estimator. Retreat used eight zero-arm/open-gripper steps followed by 16 normalized vertical/open-gripper steps, then context reset. The common budgets were 480 steps for ArrangeTea and 256 for CuttingToolSelection, including scripted retreat actions.

No earlier complete rewind anchors were available: `oracle_rewind` is empty. The executed allocation was therefore **160, not the earlier proposed 180**. There is no oracle-rewind outcome or upper-bound claim. SAFE alarms did not initiate these interventions.

The engineering and scientific JSON exports were supplied by the user from the cluster on 12 October 2026 and are preserved under `evidence/xr1_operator_screen_20261012/`. Their provenance file records hashes and implementation commit `cf0b1024`. The scientific export identifies source ledger

`85ddb0ed862005af77f3f0728a56602d5044cffd79b508edcf174ebdfa38b98e`.

Raw cluster payloads and the exact operator-run directory were not supplied locally; this report records the exported audit rather than claiming an independent local replay audit.

8.2 Observed scientific results

Table 17 uses equal task weighting after averaging both continuation seeds within each snapshot. Completion means the frozen active predicate became true at some point within the budget. Durable completion means it remained true at the branch endpoint, not that the next stage or full task completed.

Table 17: Completed operator screen. All values are percentages; gains are percentage points relative to continuation. The intervals are conditional developmental paired-bootstrap intervals.

Operator	Complete	Retained	Gain	Gain 95% interval
Continue	**22.5**	**20.0**	0.0	Reference
History reset + replan	17.5	17.5	−5.0	$[-15.0, 0.0]$
Stage retry	17.5	17.5	−5.0	$[-15.0, 0.0]$
Retreat + reset + replan	17.5	15.0	−5.0	$[-12.5, 0.0]$

ArrangeTea completion was 45% for continuation and 35% for each intervention; retained completion was 40%, 35%, 35%, and 30%, respectively. CuttingToolSelection completion and retained completion were zero in all four arms. Full-task success was zero throughout. None of the three interventions produced positive mean completion advantage in any parent, and none passed the registered scientific gate.

After averaging the two seeds, oracle selection of the best operator per snapshot reached 22.5%, exactly matching continuation. Oracle-routing-minus-best-fixed was zero, with bootstrap interval $[0, 0]$. This means no snapshot in the observed two-seed support had a positive mean completion gain from an intervention. The degenerate resampling interval reflects the observed finite support; it is not proof of zero routing value across unseen seeds, parents, or tasks.

The exported predicate-regression rates were 60% for continuation, 60% for history reset, 52.5% for stage retry, and 57.5% for retreat. In ArrangeTea, continuation's flag rate was 100%. The implemented flag detects an initially true predicate becoming false; without predicate-level inspection it must not be equated with irreversible loss of an ordered completed stage. Some relational predicates may normally change during legitimate progress. Declared safety events were zero in every arm, which is a recorded endpoint rather than a comprehensive collision-safety certification. Mean policy calls were 23 for continuation, history reset, and stage retry and 22 for retreat; mean suffix execution times were approximately 39.8–40.0 seconds across arms.

If the result is negative. The registered decision is `mechanism_redesign_or_earlier_snapshot_study`. Continuation is the best fixed operator; no operator is recommended and the promising-operator set is empty. Preserve this screen, stop scaling its operator/snapshot combination, and do not train an operator router, reopen local candidate reranking, or claim SAFE-triggered recovery benefit. The small two-task result does not establish that recovery in general is ineffective or that timing caused the failure.

8.3 Next proposed experiment: common-stage-entry timing diagnostic

Test whether these bounded interventions help before the existing stage boundary, which follows two genuine policy inferences. Use the same 10 parents as a *development-only* diagnostic, not as a new confirmation set. First replay each original nominal parent using its frozen environment and policy sampling provenance, capture a genuine complete simulator-policy state at the first eligible inference boundary of the relevant target-stage visit, and audit the original prefix-2 reference. A matching task name or reset seed alone does not establish reproduction. Missing states must be recollected; an existing late snapshot cannot be relabeled as earlier.

From each common stage-entry state, freeze two nominal continuation-seed schedules. For each seed, compare the three interventions applied immediately, after one genuine policy inference, or after two genuine policy inferences. Every delayed arm follows the identical paired nominal prefix before intervention. Use one shared continuation trajectory per parent and seed; it is the control for all three timings. This gives

$$10 \text{ parents} \times 2 \text{ seeds} \times (1 \text{ continuation} + 3 \text{ operators} \times 3 \text{ timings}) = \boxed{200 \text{ unique scientific trajectories}}.$$

An implementation that repeats the control in every timing cell allocates 240 runs, but those repeated controls are not independent evidence. Exact sentinel repeats are additional engineering executions.

All regimes end at the *same absolute environment-step endpoint from stage entry*: 480 steps for ArrangeTea and 256 for CuttingToolSelection. Waiting and scripted retreat consume that budget. Do not reuse the old prefix-2 suffix outcomes as matched late controls, since their endpoints and seed schedules differ. Freeze per-request seeds by the common nominal/request schedule before collection, and log timing eligibility, intervention execution, stage transitions, full-task success, and errors. If a target completes before a delayed intervention or the regime becomes ineligible, keep that assignment and use a frozen no-intervention fallback; do not drop it from the analysis.

The primary contrast is each operator/timing regime minus its paired continuation control. The timing contrast compares the *same* operator at early and delayed application from the shared base and endpoint. Report active-predicate completion, endpoint retention, full-task success, audited ordered-prefix regression, declared safety, calls, and latency separately. Keep tasks fixed in the paired parent bootstrap, retain all regimes and seeds from each sampled parent, and label the estimates exploratory.

Engineering gate and decision. Run a two-parent sentinel, one parent per task, to validate stage-entry replay, complete-state restoration, pre-intervention prefix equality, seed accounting, timing eligibility, shared controls, common endpoints, and exact repeats. If it passes, collect all 10 parents regardless of sentinel efficacy. A timing/operator cell is a candidate for *fresh-parent validation* only if it passes the existing practical gain and harm guardrails, including at least 10 percentage points of completion and retention gain, positive task signs, and at least three benefiting parents. Choosing among nine cells is development selection, not confirmation. If useful intervention value is still absent, redesign the executor or scope the work to the negative causal benchmark; do not run another timing or detector sweep on these parents.

Implementation status. The completed operator collector and analyzer are implemented. This timing study is a proposed protocol, documented in `docs/xr1_earlier_intervention_timing_study.md`; a common-stage-entry replay collector and dynamic-regime analyzer still need implementation and engineering validation. The existing `run_xr1_operator_screen.sh` launcher reuses the frozen late snapshots and cannot run this design.

9 Phase 4: Construct intervention-specific recoverability surfaces

Scientific purpose. Create the paper's central causal dataset by branching multiple recovery operators at multiple prespecified times from identical long-horizon states, then estimate where each intervention helps, harms, or becomes too late.

9.1 Scientific hypotheses

This phase tests three hypotheses:

1. $\Delta_k(h_t, d)$ depends on active stage, failure mode, and recovery operator;
2. recoverability can be non-monotonic because of self-correction and stage transitions;
3. terminal failure confidence is an imperfect ranking of intervention advantage and time remaining before the last useful intervention.

The contribution is not the generic observation that long delays are bad. The distinctive object is a dense, paired, multi-operator surface measured from identical simulator-policy states. This differs from applying one repair at one trigger and sweeping only execution latency.

9.2 What must be implemented

Current dependency. Both the local-candidate screen and the subsequent 160-suffix operator screen failed. Phase 4 remains locked: continuation was best and oracle routing added zero completion gain. A small common-stage-entry timing diagnostic is proposed first (Section 8.3); any selected operator/timing rule requires fresh-parent validation before a broad surface grid.

Operator interface. Define a common recovery operator API with explicit `prepare`, `act`, and `end` methods plus a cost record. Phase 4 must use only the deployable recovery mechanism or mechanisms that showed headroom in Phase 3B; it must not populate an expensive grid with failed arms. The available registry should include:

- `continue`: no intervention;
- `history_reset_replan`: clear policy image/state context and replan; a cache-only flush is redundant at empty-cache inference boundaries;
- `stage_retry`: re-prompt with the current atomic goal; the completed screen used supervisor-supplied goals, while a deployable causal stage tracker remains to be validated;
- `retreat_reset_replan`: execute a frozen bounded retreat/open-gripper/reset sequence, then replan;
- no candidate-selection arm under the current evidence; such an arm could return only after a materially different proposal family establishes new oracle headroom under a separate frozen screen;
- `env_to_last_good`: optional simulator-oracle diagnostic requiring genuine earlier anchors; none were executed in the operator screen and its success is not a guaranteed upper bound.

State and delay grid. Use three detector-independent intervention landmarks—early, middle, and late—within the active stage as the primary design. Generate all three from the same stage-entry

snapshot: restore that state, follow the same frozen nominal trajectory to the assigned intervention time, apply the operator, and evaluate at a common absolute endpoint measured from stage entry. This identifies a dynamic intervention-regime contrast under shared pre-intervention history. Merely comparing unrelated early and late snapshots does not prove that delay caused the difference.

For trigger-conditioned latency analysis, apply delays such as $\mathcal{D} = \{0, 1, 2, 4\}$ genuine inferences only on a smaller secondary subset. Keeping this factor out of the primary grid prevents a combinatorial expansion. Cached environment actions are not independent warning opportunities. Equal post-intervention suffix length may be reported as a secondary compute-matched view, but it must not replace the common-endpoint primary analysis because it grants later interventions more total task time.

Analysis tooling. Add a surface builder that estimates R_k, Δ_k, uncertainty, task/stage support, censoring, and the operational deadline in Equation (4). It must preserve the difference between natural and injected failures and between oracle and deployable context.

Proposed modules. Add `robocasa/recovery/operators.py`[new], `robocasa/recovery/collect_recoverability_surface.py`[new], and `robocasa/recovery/safe/analyze_recoverability_surface.py`[new]. Reuse ordered stage evaluation from `robocasa/recovery/subtask_eval.py` and state restoration from Phase 2.

9.3 Experimental design

Phase 4A: development, support, and variance. Select eight composite tasks before inspecting surface outcomes, stratified by horizon, semantic-stage count, baseline success, and failure support. Reserve whole task or scene identities rather than descendants of development parents. A planning design is eight tasks, 12 parent identities per task, three common-base intervention times, four arms including `continue`, and two predeclared continuation-seed schedules, yielding $8{\times}12{\times}3{\times}4{\times}2 = 2{,}304$ suffix executions before censoring. Retain only operators supported by the higher-level operator screen and reduce the arm count rather than padding the study with failed mechanisms. On a stratified subset, use additional continuation seeds to estimate seed variance.

Open Phase 4A arm identities only for support, variance, operator, and threshold selection. Freeze the retained arms, endpoints, and allocation rule afterward. Expand to approximately 12 tasks only through a separate **Phase 4B** confirmation with sealed parents and whole task or scene identities. The same 2,304 branches cannot serve as both method-selection and confirmatory evidence.

For a single snapshot and a single seed, the outcome is a realized finite-seed utility, not a probability. Define the empirical operator value as

$$\hat{R}_k(h_t, d) = \mathrm{mean}_{m=1,\ldots,M}\, Y_{k,m}(h_t, d), \tag{15}$$

where the M continuation seeds are declared before branching. Population-level recovery probabilities average these finite-seed outcomes across prespecified snapshots and tasks. Reports must show M, seed variance, and sensitivity to the continuation schedule rather than treating one suffix as an exact-state probability estimate.

Natural failures and successful controls are primary. Injected actuation noise, visual occlusion, distractors, or state perturbations may be added as a separate stress-test stratum. They must not be pooled into natural failure prevalence or the primary deployment estimand.

9.4 Evaluation and scientific tests

1. **Recoverability curves.** Plot $R_k(h_t, d)$ and $\Delta_k(h_t, d)$ by task, stage, operator, and delay with task/parent/snapshot uncertainty.

2. **Deadline.** Estimate the last supported delay at which the lower confidence bound of Δ_k exceeds a positive margin. Report grid-resolution uncertainty rather than pretending to know a continuous exact time.
3. **Integrated value.** Summarize the area under the supported Δ_k curve, weighting tasks equally and reporting the supported risk set.
4. **Heterogeneity.** Fit an interpretable mixed-effects or hierarchical model with task/stage/failure-type effects, and retain raw task-level curves.
5. **Failure probability versus value.** Compare SAFE score ranking with realized Δ_k, best-operator identity, and deadline. Use within-snapshot and task-level analyses.
6. **Rescue and harm.** Report rescue among failed nominal suffixes and corruption among successful nominal suffixes. Do not summarize only average success.
7. **Downstream retention.** Report whether an apparent rescue persists through the next stage landmark and to final task outcome when supported.
8. **Risk sets.** Publish the number of states still eligible at every later landmark; stages that naturally terminate before the late intervention cannot silently disappear from denominators.

Positive result. Phase 4A supports several task-stage strata in which at least one deployable operator has positive paired advantage, the best operator or deadline varies across states, and failure probability does not perfectly rank causal recovery value. Support is distributed across tasks rather than one favorable stage, immediate gains persist downstream, and the resulting variance estimate makes a sealed Phase 4B feasible. Only Phase 4B can confirm the frozen surface and justify the corresponding general claim.

If the result is negative. If only environment rewind helps, the current deployable recovery operators are insufficient. If all deployable Δ_k values are near zero, narrow the project to a diagnostic benchmark or change the recovery mechanism. If only one task contributes signal, do not claim a general long-horizon recoverability surface.

9.5 Validity controls

Branch-state selection must be detector-independent. Later prefixes require explicit risk-set denominators because short stages have already completed or failed. Common randomness must be shared until treatment divergence. Oracle simulator stage, oracle failure onset, and environment rewind must be shown separately from the deployable controller.

9.6 Required outputs

The dataset release should contain separate Phase 4A and Phase 4B manifests, full snapshot hashes, branch provenance, natural/injected labels, common-base and continuation-randomness schedules, common absolute endpoints, support/censoring and risk-set tables, recoverability curves, deadline estimates, downstream retention, rescue/harm plots, and a failure taxonomy with representative videos.

10 Phase 5: Train only the recovery-value model justified by causal headroom

Scientific purpose. Learn a candidate critic or operator router only when the preceding branches show that state-conditioned selection can improve over continuation and over a simpler fixed recovery policy. Validate predicted causal advantage on sealed parents and held-out task identities.

10.1 Mechanism gate before modeling

The learned model is determined by the causal evidence, not chosen in advance:

- mixed local chunks with oracle benefit over random and nominal → candidate-conditioned critic;
- heterogeneous higher-level operator utility with routing headroom over the best fixed operator → operator router;
- one uniformly useful replan or recovery operator → fixed intervention policy, with no ranking critic;
- no deployable causal headroom → no learned controller or flow guidance.

This rule prevents fitting a model simply because candidate rows exist. SAFE remains a frozen baseline and optional input; it is not assumed to encode recovery value.

Current route. Phase 3B blocked the candidate critic, and the completed 160-suffix operator screen also blocked an operator router or fixed executor at the current states: continuation was best, no parent benefited on average, and oracle routing gain was zero. Phase 5 remains locked. The proposed timing diagnostic can nominate an operator/timing cell for fresh-parent validation, but it cannot itself establish a confirmed learned recovery target.

10.2 Scientific model

For candidate action chunk $a^{1:H}$ and operator k, learn

$$Q_\phi(h_t, a^{1:H}, k) \approx \Pr(Y = 1 \mid \mathrm{do}(a^{1:H}, k), h_t). \tag{16}$$

The decision score is predicted advantage relative to continuation,

$$\hat{\Delta}_{\phi,k} = Q_\phi(h_t, a^{1:H}, k) - \hat{R}_{\mathrm{continue}}(h_t) - \lambda C_k - \mu H_k. \tag{17}$$

The critic must see the candidate or operator. A model that consumes only the current history can predict failure risk or state recoverability, but cannot rank two actions proposed from the same state. If the higher-level operator screen supports only a fixed operator, state-only modeling may be used for an intervention gate but must not be described as action or operator ranking.

10.3 What must be implemented

Inputs and ablations. Preserve separate input arms:

1. task/stage/time and progress only;
2. terminal SAFE scalar;

3. raw candidate SAFE/action token;
4. decoded action statistics;
5. causal proprioceptive state history;
6. action token plus proprioceptive state;
7. causal visual embeddings plus state history as a separately implemented ablation;
8. full action, observation, task language, predicted stage, and progress model;
9. oracle-stage version as an upper bound.

The raw original action-feature path must remain an explicit pinned ablation when context is added. The repository's current "observation context" is flattened proprioceptive history; a visual-context claim requires a new RGB embedding path with causal timestamps and separate normalization. For held-out tasks, task and stage representations must be generated from task language or a semantic descriptor, with a declared unknown-token fallback. A closed learned task-ID catalog cannot support task-generalization claims.

Training targets. Use paired candidate/operator advantage over `continue` as the primary target. Stage completion within the frozen horizon supplies the immediate binary outcome; downstream retention, full-task success where available, progress gain, prior-stage regression, and safety cost remain distinct auxiliary targets. A practical multi-objective loss is

$$\begin{aligned}\mathcal{L}(\phi) = \mathcal{L}_{\text{BCE}} + \alpha\mathcal{L}_{\text{pairwise}} + \beta\mathcal{L}_{\text{calibration}} \\ + \gamma\mathcal{L}_{\text{harm}},\end{aligned} \tag{18}$$

with equal snapshot and task weighting. Hyperparameters must be selected on parent-grouped validation data only.

Deployable stage tracker. Train a causal stage estimator from task language, observation/state history, and optionally visual embeddings. Its outputs are active-stage distribution, transition confidence, and completed-stage prefix. Measure macro stage F1 and transition latency. Oracle predicate stage is used only for an upper bound. If the critic works only with oracle stage, it is not deployable.

Proposed modules. Add `robocasa/recovery/safe/recovery_critic.py`[new], `robocasa/recovery/safe/train_recovery_critic.py`[new], `robocasa/recovery/safe/evaluate_recovery_critic.py`[new], and `robocasa/recovery/safe/stage_tracker.py`[new]. The same interface may instantiate a candidate critic or operator router, but only the evidence-supported route is trained. Extend `robocasa/recovery/safe/xr1_best_of_k.py` only on the candidate route, with a runtime distinct from the terminal SAFE ensemble.

Tests. Test parent/snapshot split inheritance, equal-snapshot weighting, strict runtime loading, score direction, candidate permutation, missing-context masking, predicted-stage versus oracle-stage paths, and exact compatibility with the Xiaomi candidate-seed interface.

10.4 Data splits and efficient model development

Split by parent before snapshots or candidates are generated. Every candidate from a snapshot stays in one split. Reserve complete task/scene identities for generalization. A practical order is:

1. fit logistic or linear probes for every input arm with one seed;

2. train one small MLP for the top three representations;
3. select one representation, objective, and regularization on validation parents;
4. refit only the top two frozen arms with three initialization seeds;
5. open sealed parents and held-out tasks once.

The effective sample size is the number of independent snapshots, parents, and tasks, not candidate rows. Large neural architectures are unlikely to be justified by the initial corpus.

10.5 Evaluation

Table 18: Recovery-critic endpoints.

Endpoint	Definition	Scientific interpretation
Within-snapshot pairwise ROC	Probability that the model ranks a higher-advantage candidate or operator above a lower-advantage alternative from the same state.	Decision-specific discrimination.
Top-1 recovery	Recovery rate of the selected candidate or operator.	Direct offline selection utility.
Oracle regret	Oracle-best finite-set outcome minus model-selected outcome.	Remaining decision headroom, including finite-seed maximization optimism.
Calibration/Brier score	Agreement between predicted and observed success/advantage.	Whether thresholds and uncertainty are meaningful.
Successful-control harm	Harm rate of the selected recovery rule versus its strongest matched control.	Safety of intervention selection.
Held-out task effect	Task-macro selected method minus strongest matched control on unseen identities.	Generalization beyond memorized task offsets.

Bootstrap tasks, then parents, then snapshots; retain all candidates from a sampled snapshot. Report per-task effects and uncertainty across model seeds separately.

Positive result. On the candidate route, the sealed-test within-snapshot pairwise ROC is at least 0.65 with lower interval above 0.50, top-1 recovery exceeds random $K = 4$ and nominal continuation by a practically meaningful frozen margin, task signs are broadly positive, and harm rises by no more than five points. On the operator route, the learned router must outperform the strongest frozen fixed deployable operator, not merely random operator choice. These are planning gates; either result establishes offline causal actionability, not yet prospective control.

If the result is negative. Do not guide or control the policy. Diagnose target-horizon mismatch, insufficient proposal diversity, stage leakage, context overfit, and the gap between stage completion and full-task success. A strong recoverability benchmark and failure-risk-versus-utility negative finding may remain publishable even when the critic fails.

10.6 Required outputs

Release the mechanism-selection record, grouped split manifests, model checkpoints and runtime bundle, feature/normalization provenance, candidate- or operator-level sealed predictions, calibration curves, top-1 and regret tables, stage-tracker confusion and latency, oracle-versus-predicted-stage gap, and representative model error cases.

11 Phase 6: Prospective fixed-trigger recovery-decision pilot

Scientific purpose. Isolate the selected recovery mechanism from trigger quality. Under identical detector-independent snapshots, test whether the frozen candidate critic, operator router, or simple fixed recovery policy produces greater causal utility than its strongest matched non-oracle control.

Current status. This phase remains conditional and has not been run. The completed candidate and operator screens provide no useful executor or selection headroom at the studied states. Resolve the bounded timing/operator capability question and validate any selected mechanism on fresh parents before proceeding. SAFE-triggered recovery benefit has not been tested.

11.1 Scientific hypothesis

For an identical observation and a frozen decision set, let $A^\star$ denote the learned or selected recovery rule and A^c its strongest matched control. Then

$$H_6: \quad \Delta_{\text{decision}} = \Pr(Y_{\text{stage}} = 1 \mid A^\star) - \Pr(Y_{\text{stage}} = 1 \mid A^c) > 0. \tag{19}$$

The fixed trigger deliberately removes detector timing from this experiment. If the chosen mechanism fails here, a later end-to-end result cannot be repaired by claiming that the trigger was responsible.

11.2 What must be implemented

On the candidate route, extend the current Xiaomi best-of-K policy path so candidate scoring can be selected from either the frozen terminal SAFE ensemble or the frozen recovery critic. Log the complete candidate set before selection, including action chunks, raw features, critic scores, terminal SAFE scores, candidate seeds, selected index, diversity, tie status, and runtime. On the operator route, log the complete eligible operator set, router values, selected operator, fallback, stage confidence, cost, and runtime. On the fixed-operator route, do not add a learned selector.

All $K = 4$ arms for an identity must receive identical first candidate sets. After different actions are executed, later states and candidate sets will diverge; analysis remains paired by initial task/environment identity and follows intention to treat.

Existing modules. Reuse `robocasa/recovery/xiaomi_robotics_1_policy.py`, `robocasa/recovery/safe/xr1_best_of_k.py`, `robocasa/recovery/recovery_rollout.py`, and `robocasa/recovery/evaluate_recovery_benchmark.py`. The current stable centered SAFE score is a ranking probe, not the calibrated alarm score, and must remain labeled accordingly.

Proposed additions. On the candidate route, add a scorer protocol field and strict runtime loader for the recovery critic. On the operator route, add a frozen eligible-operator registry, router runtime, and matched operator-set audit. Test that paired arms receive identical candidates or eligible operators before selection, that choices differ only through the declared rule, and that ordinary high-level inference remains unchanged outside recovery.

11.3 Pilot design

Use the five development tasks and 10 new parent identities per task. Trigger recovery at a fixed detector-independent stage prefix or simulator-oracle snapshot shared across arms so that only the recovery decision changes. Freeze exactly one route:

- **candidate route:** nominal $K = 1$, random $K = 4$, lowest terminal-SAFE $K = 4$, and highest predicted recovery-value $K = 4$;
- **operator route:** `continue`, the strongest fixed deployable operator, random eligible operator, and router-selected operator;
- **fixed-operator route:** `continue` versus the single supported fixed operator, with more identities allocated to the two-arm comparison rather than inventing selector ablations.

The four-arm routes produce 200 high-level arm rollouts. Compute finite-set oracle selection offline from the Phase 3B–5 corpus; do not execute it as a deployable arm.

Block and interleave assignments by task and server session. Match the initial environment seed/reset and ordinary-policy sampling family. Also match candidate seeds, recovery horizon, and the fallback seed schedule. Run arms sequentially against one stateful server or use separately validated servers with the same model and sampling behavior.

11.4 Evaluation

The primary endpoint is active-stage recovery. Full-task success, ordered progress, completed-stage regression, collision/safety events, number of policy calls, and wall-clock latency are secondary. Report:

- learned selection minus the strongest matched random and fixed controls;
- terminal SAFE candidate selection when the candidate route is active;
- generic proposal or operator benefit relative to nominal continuation;
- regret to finite-set oracle selection;
- effects conditional on causal headroom, diversity, stage, and predicted uncertainty.

Use paired task/parent bootstrap or a blocked randomization test that preserves initial identity. Errors after assignment count against the assigned arm in the primary intention-to-treat analysis; a per-protocol sensitivity may be reported separately.

Positive result. This 50-identity pilot is a screening gate, not the definitive efficacy test. On a learned-selection route, proceed when the model improves task-macro active-stage recovery over the strongest fixed deployable control by a frozen practically meaningful margin, approximately 10 percentage points, without one-task dependence or material harm. On the fixed-operator route, require benefit over continuation with the same safety guardrails. Report the paired hierarchical interval and width, but do not require this small pilot to exclude zero; Phase 7 must power and confirm the frozen primary comparison.

If the result is negative. Do not implement flow guidance. If offline selection passed but the prospective fixed-trigger test fails, preserve the discrepancy as distribution-shift or proxy-mismatch evidence. One new development cohort may be justified, but never tune on the prospective identities. If a fixed operator also fails, no learned trigger can make that executor useful.

11.5 Required outputs

Produce the interleaved allocation manifest, candidate- or operator-set equality audit, intention-to-treat sample-flow table, paired stage/task success table, latency and call accounting, error ledger, harm/regression analysis, and matched videos of rescue, harm, tie, and model-error cases.

12 Phase 7: Frozen end-to-end trigger and recovery trial

Scientific purpose. Test whether the complete deployable system improves final long-horizon task success while controlling unnecessary interventions, harm, latency, and compute. This is the primary experiment for a robotics conference paper.

Current status. This phase remains conditional and has not been run. The completed candidate and operator screens provide no useful executor or selection headroom at the studied states. Resolve the bounded timing/operator capability question and validate any selected mechanism on fresh parents before proceeding. SAFE-triggered recovery benefit has not been tested.

12.1 Scientific hypothesis

The primary hypothesis is

$$H_7: \quad \Delta_{\text{success}} = \mathrm{E}_{\text{task}}[\Pr(Y = 1 \mid A = \text{utility}) - \Pr(Y = 1 \mid A = \text{best baseline})] > 0. \tag{20}$$

The utility system uses the causal stage tracker and, only when justified by Phase 5–6, predicted $\hat{\Delta}_k$ to decide whether to intervene and which deployable operator or candidate selector to use. If the causal screen supports one fixed operator but no routing headroom, the final system uses that simpler executor. The strongest baseline is selected on development data and frozen before the final trial.

12.2 What must be implemented

Online monitor. At each genuine policy inference, collect the frozen SAFE/action feature and causal observation/state history, update the stage tracker, and compute failure-risk baselines and recovery-value predictions. Cached actions do not produce extra monitor updates.

Utility gate and optional router. Trigger only when the lower-confidence estimate of positive intervention advantage exceeds the frozen threshold and the episode budget is not exhausted. Route to $\arg\max_k \hat{\Delta}_k$ only if operator or candidate routing passed Phase 6; otherwise invoke the frozen fixed recovery operator. Use a safe fallback when model uncertainty or stage confidence is too low.

Matched executors. Use the same frozen recovery executor in the periodic, stage/time, and SAFE trigger arms so their comparison isolates trigger timing. The recovery-decision contribution has already been isolated under a fixed trigger in Phase 6.

Operating-point calibration. Calibrate alarm and utility thresholds on successful parents disjoint from final evaluation. The calibration unit is the maximum eligible score per successful parent, not an inference. For a 5% successful-parent intervention target, use at least 200 successful calibration parents; approximately 400 is preferable for a precise estimate.

Proposed modules. Add `robocasa/recovery/safe/runtime_monitor.py`[new], `robocasa/recovery/safe/recovery_router.py`[new], and `robocasa/recovery/evaluate_online_recovery.py`[new]. Integrate through the policy hooks and inference-record path already present in `robocasa/recovery/xiaomi_robotics_1_policy.py`.

Runtime tests. Test that monitoring is causal, one record is consumed per genuine inference, thresholds are immutable, successful-parent intervention count obeys budget, intervention modes use matched horizons/calls, router fallbacks are deterministic, and all assignments survive resume/partial-run handling.

12.3 Experimental arms

The final deployable comparison should include:

1. no intervention;
2. invocation-matched periodic recovery;
3. task/stage-conditioned elapsed-time trigger;
4. frozen terminal SAFE or a confirmed *predicted-stage* SAFE trigger; predicate-stage SAFE remains an oracle arm;
5. proposed recovery-value trigger and critic-selected recovery.

Add oracle failure-onset timing, oracle stage, oracle best operator, and environment rewind only as separate upper bounds. A random trigger matched to the full method's alarm count and timing distribution is a useful secondary control.

Use at least 12 composite tasks, including current development tasks and at least four held-out task or scene identities. Power one primary two-arm comparison—the full utility system versus the strongest frozen deployable baseline. Run the additional diagnostic trigger arms on a smaller predeclared cohort or increase the total budget; do not dilute power across every ablation. A planning floor is 30-50 unique parent identities per task and main arm. Final sample size must be frozen through a blinded hierarchical power simulation using pilot baseline rate, task/session variance, and paired discordance rather than the observed treatment effect.

Randomize within task, layout, seed/reset family, and server-session block. Interleave arms within each allocation so model/server drift cannot align with treatment. Match initial environment and ordinary-policy seeds; analyze intention to treat after trajectories diverge.

12.4 Primary and secondary evaluation

Table 19: Final prospective endpoints.

Endpoint	Definition	Role
Final task success	Equal-task absolute success difference, utility system minus strongest frozen baseline.	Primary paper endpoint.
Rescue	Baseline fails and utility system succeeds on matched identity.	Positive embodied benefit.
Harm	Baseline succeeds and utility system fails.	Intervention safety.
Successful-parent intervention	Any intervention during a would-be successful parent.	Episode-level false-intervention burden.
Deadline lead	Trigger time relative to the operator-specific last useful delay; misses charged at horizon.	Actionable timing.
Router regret	Oracle branch value minus chosen operator/candidate value.	Decision quality.
Cost	Alarms, interventions, candidates, model calls, NFE, GPU time, and wall-clock latency.	Deployment feasibility and matched-budget validity.
Safety	Collision, action saturation, oscillation, prior-stage regression, and repeated-recovery failure.	Guardrail outcome.

"Would-be successful" is counterfactual and cannot be inferred from the intervened trajectory alone. Estimate it from the paired no-intervention clone with the same initial identity and common randomness up to divergence; report successful-parent intervention and harm only on this matched population, with discordant pairs retained.

Report task-level effects, benefit/harm discordance, blocked randomization inference, task/parent hierarchical bootstrap, and mixed-effects sensitivity. Use one primary comparison and adjust secondary model/arm families with a prespecified hierarchical order or Holm correction.

Artificially add execution delays $d \in \{0, 1, 2, 4\}$ genuine inferences in a secondary latency sweep. This tests whether the learned gate predicts remaining recovery time without making monotonicity an assumption.

Positive result. The lower 95% hierarchical confidence bound on task-success lift is above zero, the absolute gain is at least five percentage points, the sign is positive in most task blocks, and benefit is not explained by one task, seed family, or server session. Successful-parent harm and intervention burden remain within the frozen budget, and latency/cost are reported at matched conditions. This result supports the deployable recovery-system claim.

If the result is negative. Do not claim online recovery utility. Use the oracle decomposition: if oracle recovery helps but utility triggering does not, timing or value prediction is the bottleneck; if alarms are early but no operator helps, recovery execution is the bottleneck; if successful parents are harmed, the intervention policy is unsafe; if only environment rewind helps, the result is simulator-oracle only.

12.5 Power interpretation

The current Xiaomi timing study remained imprecise with 380 failed parents. If a 0.05-horizon timing effect remains a primary claim, approximately 400-600 failures may be required before task clustering, with 500-700 a defensible planning range after heterogeneity inflation. Similarly, an improvement from approximately 12% to 20% task success needs roughly 330 parents per arm before clustering; server/task heterogeneity can raise the requirement to approximately 500-650 per main arm. These are planning calculations and must be replaced by a frozen simulation using the observed blinded-pilot design variance.

12.6 Required outputs

Release assignment, started/completed/error/analyzed flow; immutable outcome manifest; primary randomization and bootstrap results; per-task success; rescue/harm; calibration; deadline and latency; router regret; compute/safety tables; and predeclared representative videos, including no-headroom and harmful cases.

13 Phase 8: Add differentiable recovery-value guidance to the flow policy

Scientific purpose. Attempt an algorithmic efficiency extension only after discrete critic reranking proves prospective actionability. Test whether a noise-step-conditioned critic can guide one action-generation trajectory as effectively as sampling and ranking four complete candidates.

Current status. This phase remains conditional and has not been run. The completed candidate and operator screens provide no useful executor or selection headroom at the studied states. Resolve the bounded timing/operator capability question and validate any selected mechanism on fresh parents before proceeding. SAFE-triggered recovery benefit has not been tested.

13.1 Scientific hypothesis and relation to prior work

Failure-informed or critic-guided diffusion/flow improvement is already an active area [6, 7, 8, 3, 9]. In particular, Guided Action Flow directly studies test-time action-gradient guidance for a frozen flow-matching VLA. Therefore, "using a critic to guide flow" or "SAFE plus a critic" is not by itself a sufficient novelty claim. Within this project, guidance is valuable only as a conditional efficiency extension: it must use the causal recovery-value target in Equation (3), operate only when the utility gate identifies a recoverable state, and achieve the same or better embodied recovery with lower matched cost.

Let z_τ be the noisy action latent and $v_\theta(z_\tau, h_t, \tau)$ the frozen policy velocity. A generic guided form is

$$v_{\text{guided}} = v_\theta + s(\tau)\,\Pi(\nabla_{z_\tau} Q_\phi(h_t, z_\tau, \tau)), \tag{21}$$

where $s(\tau)$ is a frozen scale schedule and Π clips or projects the gradient to preserve action bounds. The exact sign and parameterization must match the Xiaomi flow implementation; Equation (21) is a design template, not evidence that the current detached client score is differentiable.

13.2 What must be implemented

1. Expose the noisy action/token latent and flow integration step inside the Xiaomi model-side generation loop.
2. Train or distill a critic conditioned on z_τ and τ, using the same counterfactual recovery outcomes as Phase 5.
3. Execute the critic inside the model server and preserve the original unguided action-only and feature-return behavior when guidance is disabled.
4. Freeze guidance sign, scale schedule, integration-step window, suffix scope, gradient norm clipping, action deviation limit, and uncertainty fallback.
5. Log intermediate critic values, gradient norms, clipping frequency, action deviation, NFE, backward passes, GPU time, and executed outcome.

Required architectural change. The current SAFE/best-of-K scorer receives detached final candidate features on the client. True guidance requires a model-side gradient path through intermediate action latents. Implement this as an opt-in server protocol with focused tests and an unguided-equivalence smoke.

13.3 Development and locked test

Develop on `ArrangeTea` and `CuttingToolSelection`, for example with 20 fresh identities per task. Tune one small guidance-scale grid and guided integration window on development identities only. Freeze the selected setting and test on the remaining current tasks plus held-out composite tasks from Phase 7.

Compare:

1. unguided $K = 1$;
2. random $K = 4$;
3. critic-reranked $K = 4$;
4. critic-guided $K = 1$;
5. shuffled-critic or random-gradient negative control.

Provide both matched-NFE and matched-wall-clock comparisons. An apparent gain obtained by spending more backward passes or latency than $K = 4$ is not an efficiency result.

13.4 Evaluation

Primary outcomes are active-stage recovery and full-task success. Secondary outcomes are predicted versus realized Δ, prior-stage regression, collision, saturation, oscillation, action-distribution shift, NFE, model calls, memory, GPU time, and end-to-end latency. Ablate guidance scale, guided step window, suffix length, terminal SAFE risk target versus recovery-value target, and the negative gradient control.

Positive result. Guided $K = 1$ matches or exceeds critic-reranked $K = 4$ within a frozen statistical margin, uses meaningfully less inference cost or latency, and does not increase collision, saturation, oscillation, or completed-stage regression. The frozen guidance setting transfers from the two development tasks to held-out tasks.

If the result is negative. Retain the Phase 6 executor, whether a critic, router, or fixed operator. Do not present guidance as successful if it raises the critic score but not executed task outcomes, if a shuffled gradient performs similarly, or if matched-compute comparisons remove the benefit. Phase 8 is optional; the causal recoverability paper does not require it to pass.

13.5 Required outputs

Release the opt-in model/server protocol, unguided-equivalence tests, noise-step critic bundle, frozen guidance configuration, matched-compute evaluation, gradient/action diagnostics, safety analysis, and videos comparing unguided, reranked, and guided executions from matched initial identities.

14 Phase 9: Generalize across tasks, scenes, policies, and physical execution

Scientific purpose. Demonstrate that the causal recoverability finding and utility gate are not artifacts of five Xiaomi tasks, one visual distribution, privileged stage information, or simulator-only recovery.

14.1 Scientific hypotheses

The strongest external-validity hypothesis is that the recovery-value formulation transfers more reliably than a raw SAFE threshold, even when feature extractors and calibration remain policy-specific. The claims should be separated:

1. **same-task generalization**: new seed/reset identities;
2. **scene/layout generalization**: new arrangements, appearances, cameras, and distractors;
3. **task generalization**: semantic task identities excluded from critic/model selection;
4. **policy generalization**: a second VLA family under the same causal protocol;
5. **embodiment generalization**: physical robot execution without simulator predicates or state rewind.

14.2 What must be implemented

Held-out simulation. Freeze at least four unseen composite tasks or scenes. Add language paraphrases, camera/view shifts, object distractors, and layout/style changes where they test a stated representation claim. The task/stage descriptions and online tracker must be fixed before outcomes.

Here "unseen task" means excluded from critic, stage-tracker, threshold, and model-selection development; it does not imply that the underlying VLA policy never encountered related tasks during its original pretraining unless that stronger provenance is independently established.

Second policy. RLDX-1 is the natural replication candidate only after a no-SAFE official-path health check demonstrates nonzero success and ordered progress. Reproduce the same snapshot, branch, label, calibration, and intervention protocol with policy-specific features and checkpoints. Never average raw Xiaomi and RLDX scores or imply that absolute SAFE metrics rank the policies.

Physical validation. Choose three or four multi-stage tabletop tasks with the main conditions: no intervention, strongest deployable baseline, and full utility system. Use predicted stage from observations, not simulator predicates; define human takeover, emergency stop, collision, object drop, timeout, retry, and exclusion rules before trials. Aim for 15-20 trials per task and main condition, or at least approximately 90 controlled recovery trials total if hardware time is limited.

Robot access, safety approval, calibrated sensing, reset procedures, and an operator-independent logging stack are external dependencies not provided by the current repository. The physical phase starts only after these prerequisites and the simulator safety gate are complete.

Policy health gate. Before any second-policy failure collection, require a model-loaded server, listening port, live GPU process, correct task list, ordinary no-SAFE success/progress, and a bounded smoke. Stop repeated zero-success or no-progress collection rather than creating a scientifically unusable failure corpus.

14.3 Evaluation

For each external axis, report task-macro task-success lift, critic ranking, calibration, intervention rate, deadline error, router regret, rescue, harm, latency, and safety. Report zero-shot critic transfer, feature-adapter transfer, and policy-specific critic retraining as separate settings.

On physical trials, use trial-level uncertainty and report all takeovers and technical failures. Manual stage or failure-onset annotations may be used for evaluation when annotators are blinded to alarms, but not for online decision making. Measure stage-estimator accuracy and transition delay independently.

Positive result. The prespecified external endpoint retains a positive effect with lower uncertainty bound above zero, no severe calibration drift, and no dependence on one task or policy subgroup. The deployable stage tracker supports the result, and gains are not obtained by increasing intervention count or compute beyond the matched baseline. A strong CoRL/RSS submission should pass at least two external axes, preferably held-out tasks plus a second policy or physical evidence.

If the result is negative. Scope the claim to Xiaomi RoboCasa365. Identify whether the gap arises from representation, calibration, stage tracking, proposal support, critic ranking, recovery execution, or embodiment. A transparent same-policy result may remain appropriate for ICRA or WACV, but it cannot support policy-agnostic or real-robot language.

14.4 Venue interpretation

For CoRL, the target package is the causal benchmark, learned utility gate, held-out task/scene evaluation, prospective closed-loop gain, and preferably another policy or robot. RSS requires especially clean causal identification, latency/safety analysis, and compelling physical-system evidence. ICRA can support a narrower but complete integrated system with broad RoboCasa365 simulation and a smaller robot study. CVPR/ICCV requires the visual/action representation or guidance method to become central, with cross-camera/object/scene and multiple-encoder experiments. WACV is a strong route for a released causal benchmark and evaluation audit with a useful learned critic.

14.5 Required outputs

Provide frozen external identities; second-policy positive-control and replication reports; physical trial and safety logs; all videos and exclusions; domain-shift and stage-tracker ablations; policy-specific calibration; and a claim table that states exactly which external axes passed.

15 Cross-phase statistical and validity plan

15.1 Confirmatory hierarchy

The report should use sequential gates rather than treating every endpoint as independently confirmatory:

1. frozen stage-aware ranking transfers to new parents;
2. complete snapshot replay makes counterfactual branches valid;
3. a four-snapshot engineering sentinel validates long-horizon continuation, after which the full 20-snapshot screen determines whether local candidates contain operational headroom;
4. candidate or higher-level operator sets contain recovery headroom over both random or fixed selection and nominal continuation;
5. a recovery-value critic ranks candidate utility on sealed data;
6. critic selection works prospectively under a fixed trigger;
7. the complete utility system improves final task success;
8. external validity and flow guidance are tested only after the core system passes.

A failed detector gate does not prohibit studying recoverability with time, progress, and observable context. A failed oracle-opportunity or critic-actionability gate does prohibit claiming a useful local recovery critic or gradient guide.

15.2 Alarm and timing definitions

For failed unit u, let H_u be the frozen normalization horizon and τ_u the first alarm. Define miss-adjusted detection time

$$T_u = \min(\tau_u/H_u, 1) \qquad \text{if detected,} \tag{22}$$

$$T_u = 1 \qquad \text{if missed.} \tag{23}$$

The primary timing contrast is $\mathrm{E}[T^{\mathrm{SAFE}} - T^{\mathrm{time}}]$; negative favors SAFE. Conditional-on-detection time is secondary because a method could appear early by detecting only a few easy failures.

For $q \in \{0.10, 0.25, 0.50\}$, early-landmark recall is

$$\Pr(\tau_u/H_u \le q \mid u \text{ fails}). \tag{24}$$

The successful-parent false-intervention rate is the probability of at least one alarm/intervention over an entire paired would-be successful parent. A treated rollout alone does not reveal whether it would have succeeded without intervention; this outcome requires a matched no-intervention clone or randomized arm under the same initial identity.

15.3 Calibration

For successful-parent maximum scores $Z_1, \ldots, Z_n$ and target α, a split-conformal upper threshold uses order statistic

$$k = \lceil (n+1)(1-\alpha) \rceil. \tag{25}$$

If $k > n$, place the threshold above the observed maximum and label the operating point unresolved. Zero empirical FPR with too few calibration successes is not evidence of precise control.

Calibration requirements are:

- successful parent is the calibration unit;
- score aggregation, tie handling, task weighting, threshold rule, and target α are frozen;
- normalization is fitted on training, not calibration or test;
- calibration and final evaluation identities are disjoint;
- report calibration count, order-statistic rank, realized FPR, and exact or Wilson interval;
- evaluate drift by task/layout/policy without updating the final-test threshold.

15.4 Randomization and paired inference

Intervention assignments should be generated before outcomes and blocked by task, layout/style, seed/reset family, and server/GPU session. Arms must be interleaved within server sessions. The primary analysis is intention to treat: differential runtime or model errors count against the assigned arm according to the frozen attrition rule.

For exactly paired initial identities, report benefit and harm discordance:

$$B = \Pr(Y^{\text{method}} = 1, Y^{\text{control}} = 0), \tag{26}$$

$$H = \Pr(Y^{\text{method}} = 0, Y^{\text{control}} = 1), \tag{27}$$

$$\Delta_{\text{paired}} = B - H. \tag{28}$$

Use blocked randomization/permutation inference as primary where assignment supports it, task-then-parent hierarchical bootstrap as a transparent interval, and mixed-effects logistic regression only as a sensitivity analysis.

15.5 Bootstrap hierarchy

For detector comparisons on identical parents:

1. resample tasks if claiming task-population generalization;
2. resample parents within each selected task;
3. retain every stage, prefix, snapshot, and candidate from the sampled parent;
4. recompute the complete metric and paired method contrast.

For candidate studies, snapshots are nested within parents and candidate branches remain paired within snapshot. Three neural initialization seeds measure fitting instability and do not replace task or parent uncertainty.

15.6 Opportunity-screen interpretation

Phase 3B-0 was an engineering sentinel with no scientific efficacy stop rule. Phase 3B-1 used 20 snapshots nested within 10 parents and two tasks; task–parent–snapshot resampling and raw task signs governed interpretation. Candidate seeds and continuation seeds were frozen and logged separately. The completed screen found one mixed snapshot from one ArrangeTea parent, task-macro oracle-minus-random of 0.0125, oracle-minus-nominal of 0.0500, and zero full-task success. The corresponding hierarchical bootstrap intervals were $[0, 0.05]$ for oracle-minus-random and $[0, 0.20]$ for oracle-minus-nominal. These cohort-specific estimates fail the registered continuation thresholds and do not justify a five-task candidate screen.

Oracle-best-of-K is a finite-seed upper bound with maximization optimism. Report oracle-minus-random and oracle-minus-nominal together: oracle superiority to random establishes rankable variation, while superiority to nominal establishes operational headroom. A case-control screen estimates conditional opportunity, not natural failure prevalence, intervention burden, or population task success.

Stage completion, next-stage durability, full-task success, prior-stage regression, and safety termination are distinct outcomes. A generic progress score can diagnose mechanism but cannot be promoted to the primary endpoint after a completion gate fails. Likewise, differences between independently sampled early and late states do not identify a causal effect of intervention delay; the common-stage-entry dynamic-regime construction in Phase 4 is required for the "earlier intervention is easier" claim.

15.7 Completed operator screen and proposed timing diagnostic

The 160-suffix operator screen averaged two seeds within each of 20 snapshots, weighted its two tasks equally, and used a fixed-task parent-then-snapshot paired bootstrap. Point completion differences were -0.05 for every intervention; intervals were $[-0.15, 0]$ for history reset and stage retry and $[-0.125, 0]$ for retreat. These intervals include zero and do not establish a general harmful effect. Oracle routing gain was zero with a degenerate $[0, 0]$ resampling interval on the observed finite support; this does not bound unseen-seed or population routing value. No parent had positive mean intervention completion gain.

The proposed timing diagnostic has one common base per parent, two seeds, three operators, and three application times, plus one shared continuation per seed: 200 unique trajectories. Preserve all timings, operators, and seeds together when resampling parents within each fixed task. Shared or repeated controls do not multiply the sample size. Freeze common absolute endpoints and retain assignments that become ineligible before intervention. Selecting among nine timing/operator cells is development model selection on already opened parents and requires a fresh-parent confirmation. The old screen must not be pooled as matched late controls because its endpoints and seed schedules differ.

15.8 Power and sample size

Final N should be frozen through simulation or hierarchical bootstrap using development or blinded-pilot variance. Allowed adaptations include extending a fixed-attempt collection because predeclared support counts are unmet and blinded variance-based re-estimation. It is not acceptable to stop when ROC or success first becomes favorable, add tasks after seeing per-task effects, or alter thresholds from test alarms.

Planning calculations from current evidence suggest:

- roughly 400 failures before clustering to detect a 0.05-horizon paired timing effect under the observed variance, and approximately 500-700 after heterogeneity inflation;
- roughly 330 parents per arm before clustering for a 12%-to-20% success difference, potentially 500-650 per main arm after task/server inflation;
- at least 200 successful calibration parents for a 5% threshold, with approximately 400 preferable;
- approximately 25 successes and 25 failures per task-stage stratum for exploratory stage estimates, and closer to 50 per class for stable confirmatory estimates.

These calculations are not promises or venue rules. The final powered design should prioritize the primary two-arm comparison in Phase 7; secondary trigger and recovery ablations can use smaller frozen cohorts.

15.9 Leakage and survivor-bias controls

Table 20: Cross-phase validity controls.

Risk	Required control
Parent leakage	Assign parents before stages, windows, snapshots, or candidate augmentation. Every descendant inherits the parent split.
Snapshot leakage	Keep all operators/candidates from a snapshot together and weight snapshots, not candidate rows.
Future information	Exclude final horizon, termination reason, future predicate states, future stage completion, future features, and realized future duration from deployed inputs.
Privileged stage	Report simulator predicate stage and causal observable stage tracker separately.
Survivor bias	Publish risk-set denominator and class support at every later prefix; treat stage completion and failure as competing events in sensitivity analysis.
Outcome quota	Use case-control/outcome-quota data for ranking or feasibility only, never deployment prevalence, FPR burden, or success rate.
Feature duplication	Store one feature per genuine policy inference, not every cached action.
Restore confounding	Restore simulator, policy queues/cache, counters, instruction, tracker, and RNG; prove exact same-request/action replay.
Compute confounding	Match candidate count, recovery horizon, NFE, policy calls, or wall-clock budget according to the stated comparison and report both calls and time.
Oracle mixing	Never average oracle stage, oracle onset, environment rewind, or oracle candidate choice with deployable results.
Selective attrition	Freeze exclusion/retry rules; preserve partial manifests and technical errors; provide assigned/started/completed/invalid/analyzed flow.

15.10 Multiplicity and reporting

Declare one primary contrast per phase and one final primary contrast: full utility system versus strongest frozen non-oracle baseline on final task success. Test secondary families in a frozen hierarchy or use Holm correction. Per-task, per-stage, per-delay, and failure-type analyses remain descriptive unless explicitly powered and adjusted. Report effect sizes and intervals even when gates fail.

Gate interpretation. Pilot thresholds such as 20% mixed snapshots, 15-point oracle-over-random headroom, 10-point oracle-over-nominal headroom, pairwise ROC 0.65, or a 10-point learned-selection gain are proposed continuation gates. They protect compute and scientific focus; they are not universal statistical truths or venue requirements.

15.11 Claim ledger

Table 21: Claim-to-evidence ledger at the current starting point.

Claim	Current status	Evidence required
Xiaomi SAFE modestly improves matched-FPR classification.	Supported in frozen prospective shadow evaluation.	Preserve modest wording and artifact provenance.
Xiaomi SAFE provides reliable overall early warning.	Rejected by timing interval and early recall.	New representation plus untouched prospective timing cohort.
Full-parent stage conditioning improves early ranking.	Rejected as a general five-task claim: 0.696 development ROC fell to 0.489 in frozen identity-disjoint confirmation.	Preserve task/stage heterogeneity as diagnostic evidence; do not refit on confirmation.
Proprioceptive state context improves early stage ranking.	Unsupported: context reached 0.566 on the retrospective 85-parent holdout versus stage 0.584 and time 0.500.	A new untouched cohort would be required for a context claim, but it is not the current priority.
SAFE ranks useful action chunks.	Unsupported for the frozen local proposal family: Phase 3B oracle-best-of-four exceeded random choice by only 0.0125, mixed support occurred in 1/20 snapshots, and full-task success was zero.	A materially different proposal mechanism would need new oracle headroom before any SAFE ranking test; the current route is closed.
Recovery-value critic or router predicts causal utility.	Blocked by the negative candidate and operator screens; oracle operator routing gain was zero.	First establish operator/timing value on fresh parents; then Phase 5 sealed evaluation and Phase 6 prospective fixed-trigger benefit.
Utility triggering improves task success.	Unsupported.	Phase 7 randomized closed-loop trial.
Flow guidance improves recovery.	Unsupported.	Phase 8 model-side, matched-compute execution.
Method generalizes to unseen tasks or policies.	Unsupported.	Phase 9 with identities unseen to critic development and separate policy calibration.
Method is real-robot deployable.	Unsupported; simulator predicates/resets are currently used.	Causal stage tracker and physical intervention study.

16 Implementation map, execution order, and venue strategy

16.1 Current versus proposed code

Table 22: Implementation map. Remaining proposed items must not be described as implemented.

Phase	Current reusable code	Proposed implementation
0	SAFE schema/dataset validators; cluster runbook; frozen stage runner.	Experiment contract, allocation validator, and cross-manifest identity audit.
1	`run_stage_aware_parent_safe.py`; `stage_aware_parent_safe.py`; frozen documentation and tests.	Confirmation count/support validator; CI criterion in the frozen gate; deliberate CI-failure test.
2	`full_snapshot.py`; `counterfactual_branch.py`; Xiaomi policy `get_state/set_state`; exact replay runner, printer, combined audit, and focused tests.	Conditional task/platform extensions only; the bounded two-task gate has passed.
3	Phase 2 candidate payloads; frozen terminal SAFE scorer; ordered subtask evaluator; `analyze_phase3_candidate_opportunity.py` and tests.	Completed negative short-horizon analysis; no additional Phase 3 implementation.
3B/3B-R	Immutable training-horizon runner/analyzer; implemented `operator_screen.py`, `run_operator_screen.py`, `analyze_operator_screen.py`, launcher and tests; completed negative 140-branch candidate and 160-suffix operator results.	Common-stage-entry nominal replay and earlier-state capture; paired dynamic-regime runner, shared control and common-endpoint audit; timing analyzer. These additions remain proposed.
4	Recovery modes, ordered progress/regression trace, policy-pluggable benchmark.	Operator API, common-stage-entry dynamic-regime collector, risk-set and common-endpoint audit, and staged recoverability-surface analysis.
5	Raw action-token features; proprioceptive state history; stage-aware weighting ideas.	Evidence-selected candidate critic, operator router, or fixed utility gate plus causal stage tracker.
6	Deterministic random/lowest/highest candidate selection and mock tests.	Route-specific fixed-trigger arm runner and matched candidate/operator-set audit.
7	Genuine-inference feature records and post-failure recovery hooks.	Pre-action Xiaomi monitor, utility gate/router, intervention budget, and online randomized evaluator.
8	Detached pre-output hidden feature and client-side no-gradient scoring.	New model/server patch for noise-step critic and differentiable flow guidance.

Phase	Current reusable code	Proposed implementation
9	RLDX raw feature adapter and separate official recovery path.	Second-policy snapshot/session restoration, frozen shift wrappers, and external physical safety infrastructure.

The current Xiaomi "observation context" is flattened proprioceptive state history, not a visual embedding. The report and code should call it *state context*. A visual-context ablation requires explicit causal RGB embeddings and their own storage, normalization, and domain-shift evaluation.

16.2 Execution order

Phase 1 failed confirmation; Phase 2 passed exact replay; Phase 3 failed the short-suffix opportunity gate. Phase 3B passed engineering but found only 0.0125 oracle-over-random headroom and no full-task success. The subsequent 160-suffix higher-level operator screen also passed engineering but failed scientifically: continuation reached 22.5% completion versus 17.5% for each intervention and oracle routing gain was zero. The next proposed experiment is the 200-trajectory common-stage-entry timing diagnostic in Section 8.3. Candidate-critic and operator-router training, broad surfaces, flow guidance, and end-to-end SAFE-triggered recovery remain blocked or deferred.

The following dependencies must remain sequential:

1. exact snapshot replay before counterfactual scale;
2. the completed engineering sentinel before the completed full training-horizon screen, with no efficacy stop based on four outcomes;
3. preserve both completed negative local-candidate and operator screens before the single proposed timing diagnostic;
4. operator-over-continuation and routing-over-best-fixed opportunity before critic or router training;
5. sealed offline model gate before prospective fixed-trigger selection;
6. prospective fixed-trigger benefit before differentiable guidance;
7. development freeze before every fresh confirmation or final trial;
8. healthy ordinary-policy positive control before second-policy failure collection.

16.3 Cluster and artifact discipline

Source code lives under `/gs/fs/tga-shinoda/felid`; environments, checkpoints, datasets, rollouts, videos, logs, and reports live under `/gs/bs/tga-shinoda/felid`. Start and verify the policy server before the simulator client. Use `python -u` under background execution, unique ports and output roots, disabled W&B by default, immutable manifests, and persistent logs. Storage quota or inode errors are stop-and-inspect events. Preserve partial evidence and resume or merge rather than deleting failed blocks.

16.4 Venue completion bars

Table 23: Recommended evidence packages. These are competitive research recommendations, not official acceptance rules.

Venue	Competitive package	Main risk
CoRL	Causal benchmark; learned or demonstrably sufficient fixed utility mechanism; held-out tasks/scenes; prospective closed-loop gain; preferably second policy or physical evidence.	Five-task detector or oracle-heavy recovery appears incremental beside CheckVLA, B2FF, and recent critic-guided flow work.
RSS	Exceptionally clean exact-state causal design, deadline/latency and safety analysis, strong physical-system evidence, and direct comparison to the closest recovery systems.	Broad benchmark tables without a crisp robotics mechanism.
ICRA	Complete integrated system, broad RoboCasa365 simulation, runtime/safety analysis, strong component ablations, and several physical multi-stage tasks where feasible.	Simulator-only one-backbone evaluation may look incomplete.
CVPR/ICCV	Recoverability as a visual/action representation contribution, multiple encoders, held-out camera/object/scene shifts, and closed-loop control validating the vision method.	A primarily trigger/control paper is a scope mismatch.
WACV	Reproducible causal benchmark/application, public code/data, rigorous negative audit, critic reranking, and meaningful visual/context evaluation.	Weak fit without released artifacts or significant vision/application content.

16.5 Recommended paper contributions

1. We introduce intervention-specific recoverability curves for long-horizon VLA execution, measured through exact-prefix counterfactual branches over task stages, delays, and recovery operators.
2. We show that terminal failure likelihood is not equivalent to causal intervention value or remaining recovery time.
3. Conditional on causal headroom, we learn the simplest justified decision mechanism—candidate critic, operator router, or utility gate—and test its prospective task-success benefit while controlling unnecessary and harmful intervention.

Critic-guided flow is a fourth contribution only if Phase 8 passes matched-compute and held-out safety tests. It cannot be the core novelty because closely related VLA guidance has already been demonstrated [8, 9]. Otherwise, retain the simpler Phase 6 executor.

16.6 Immediate next actions

1. Preserve the completed Phase 1–3B results and the 160-suffix operator screen, registrations, payloads, hashes, analyses, and error history. The local report contains supplied JSON exports and explicitly records that raw cluster payloads were not independently reaudited here.
2. Inspect matched predicate traces and nominal parent provenance, including the high ArrangeTea predicate-regression flag rate; distinguish valid relational changes from loss of ordered completed-stage progress.
3. Implement and engineering-test common-stage-entry replay and capture for the same 10 development parents. Audit reproduction of the original prefix-2 reference; do not manufacture earlier states.

4. Register three application times (zero, one, or two genuine inferences), three unchanged interventions, two paired seed schedules, a shared continuation per parent/seed, common absolute endpoints, and the 200-trajectory allocation.
5. Pass a two-parent engineering sentinel, then collect the full registered support regardless of sentinel efficacy. Report all nine cells, durability, task signs, parent support, harm, cost, and timing contrasts.
6. If one operator/timing cell passes the practical gate, freeze it for fresh-parent validation. If none helps, redesign the recovery executor or scope the paper to the negative causal benchmark; do not repeat sweeps on these parents.
7. Only after fresh-parent operator benefit is established should broad recoverability surfaces, route-specific modeling, and prospective fixed-trigger pilots be reconsidered.
8. Keep the SAFE-triggered end-to-end trial deferred until a useful executor exists; it remains a separate scientific question from the completed detector shadow tests.

First research priority. The highest-value next experiment is the bounded common-stage-entry timing diagnostic, not another detector sweep or operator-router fit. The completed operator screen found no useful fixed intervention and zero routing gain. The proposed timing collector and analyzer are not yet implemented; the current operator launcher only reruns the existing frozen snapshot support.

17 References and repository anchors

References

[1] Qiao Gu, Yuanliang Ju, Shengxiang Sun, Igor Gilitschenski, Haruki Nishimura, Masha Itkina, and Florian Shkurti. SAFE: Multitask Failure Detection for Vision-Language-Action Models. In *Advances in Neural Information Processing Systems*, volume 38, 2025. DOI: 10.52202/085713-1337; arXiv: 2506.09937.

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[9] Liuhaichen Yang, Zhuang Jiang, Chenchao Sheng, and Zezhi Tang. Guided Action Flow: Q-Guided Inference for Flow-Matching Vision-Language-Action Policies. arXiv preprint arXiv:2607.02092, 2026.

17.1 Repository evidence

The scientific starting point and run procedures are documented in:

- `reports/safe_robocasa_full_report_20260804/build/main.pdf`;
- `reports/safe_robocasa_full_report_20260804/main.tex`;
- `docs/safe_xiaomi_stage_aware_parent.md`;
- `docs/xr1_safe_best_of_k_recovery.md`;
- `docs/xr1_phase3_candidate_opportunity.md`;
- `docs/cluster_experiment_runbook.md`;
- `robocasa/recovery/analyze_phase3_candidate_opportunity.py`;
- `robocasa/recovery/recovery_rollout.py`;
- `robocasa/recovery/xiaomi_robotics_1_policy.py`;
- `robocasa/recovery/safe/stage_aware_parent_safe.py`;
- `robocasa/recovery/safe/xr1_best_of_k.py`.

Literature status. SAFE and RoboCasa365 have confirmed conference publications. CheckVLA, B2FF, CycleVLA, AFIL, RoVer, Q-VGM, and Guided Action Flow are cited conservatively as arXiv preprints. Any "first" claim requires a refreshed full-text literature audit immediately before submission.